Supporting information for "The risks of dichotomising ordinal outcomes: A spatial analysis of self-rated health in Western Europe"

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Required packages

```
# # IMPORTANT:
# # Next line shoud be run once to install the latest development version from GitHub, enabling
↪  the use of HMC with dcar_leroux
# remotes::install_github("nimble-dev/nimble", subdir = "packages/nimble")

# install.packages("pacman", dep = TRUE)

pacman::p_load(sf, spdep, readxl, abind, ggplot2, RColorBrewer, patchwork,
               nimble, nimbleHMC, MCMCvis, scales, install = FALSE)

# # Loading functions for calling Nimble
#
↪  source("https://raw.githubusercontent.com/MigueBeneito/pNimble/refs/heads/main/RutinasNimble.0.2.R")
# load.leroux()
```

European Social Survey (ESS) data loading

```
# ------------------------------------------------------------ #
# Data source: European Social Survey (ESS)
#
# To replicate the analysis, please download the dataset from:
# https://ess.sikt.no/en/datafile/242aaa39-3bbb-40f5-98bf-bfb1ce53d8ef
#
# Access to the ESS data requires a free user registration.
# Once downloaded, extract the .zip file and place the extracted
# CSV file in the "data" folder before running this script.
# ------------------------------------------------------------ #

# ESS data
ESSData <- read.csv(file.path("..", "data", "ESS11e04_1.csv"))
```

```r
# Western Europe data
country <- "west"
ESSData <- ESSData[ESSData$cntry == "AT" | ESSData$cntry == "BE" |
                     ESSData$cntry == "CH" | ESSData$cntry == "DE" |
                     ESSData$cntry == "ES" | ESSData$cntry == "FR" |
                     ESSData$cntry == "IT" | ESSData$cntry == "NL" |
                     ESSData$cntry == "PT", ]

ESSData <- ESSData[ESSData$region != "ES63" & ESSData$region != "ES64" & ESSData$region != "ES70"
↪  &
                     !grepl("^FRY|^FRZ", ESSData$region) & ESSData$region != "FRM0" &
                     ESSData$region != "PT20" & ESSData$region != "PT30", ]

# Definition of the categorical covariates
# Age groups: 1 = (14, 34]; 2 = (34, 54]; 3 = (54, 100]
ESSData$agecut <- cut(ESSData$agea, c(14, 34, 54, 100))
levels(ESSData$agecut) <- c("15-34", "35-54", "55...")
# Education groups: 1 = (-1, 225]; 2 = (225, 500]; 3 = (500, 1000]
ESSData$educut <- cut(ESSData$edulvlb,c(-1, 225, 500, 1000))
levels(ESSData$educut) <- c("edulow", "edumid", "eduhigh")

# ESS cleaned
ESS <- ESSData[, c("health", "region", "gndr", "agecut", "educut",
                   "dweight", "pspwght", "anweight", "cntry",
                   "domain", "psu", "stratum", "prob")]

# Removing NA rows for age and education groups for simplicity
ESS <- ESS[!is.na(ESS$agecut), ]
ESS <- ESS[!is.na(ESS$educut), ]
str(ESS)
```

```
## 'data.frame':    17053 obs. of  13 variables:
##  $ health  : int  3 2 1 3 2 1 2 2 2 2 ...
##  $ region  : chr  "AT31" "AT22" "AT33" "AT31" ...
##  $ gndr    : int  1 2 2 2 1 2 2 2 2 2 ...
##  $ agecut  : Factor w/ 3 levels "15-34","35-54",..: 3 1 2 3 3 3 3 3 2 3 ...
##  $ educut  : Factor w/ 3 levels "edulow","edumid",..: 2 2 3 2 2 2 1 3 2 1 ...
##  $ dweight : num  1.185 0.61 1.392 0.556 0.723 ...
##  $ pspwght : num  0.393 0.325 4 0.176 1.061 ...
##  $ anweight: num  0.13 0.1076 1.3237 0.0583 0.3511 ...
##  $ cntry   : chr  "AT" "AT" "AT" "AT" ...
##  $ domain  : int  2 1 2 1 2 2 2 2 2 2 ...
##  $ psu     : int  317 128 418 295 344 373 86 3 108 314 ...
##  $ stratum : int  107 69 18 101 115 7 58 38 62 105 ...
##  $ prob    : num  0.000579 0.001124 0.000493 0.001233 0.000949 ...
```

```r
# Transforming gender and region into factors
ESS$gndr <- factor(ESS$gndr, labels = c("Man", "Woman"))

# Western Europe
ESS$region <- factor(ESS$region,
                     levels = c("AT11", "AT12", "AT13", "AT21", "AT22", "AT31", "AT32",
                                "AT33", "AT34", "BE10", "BE21", "BE22", "BE23", "BE24",
                                "BE25", "BE31", "BE32", "BE33", "BE34", "BE35", "CH01",
                                "CH02", "CH03", "CH04", "CH05", "CH06", "CH07", "DE1",
```

```
                                "DE2", "DE3", "DE4", "DE5", "DE6", "DE7", "DE8", "DE9",
                                "DEA", "DEB", "DEC", "DED", "DEE", "DEF", "DEG", "ES11",
                                "ES12", "ES13", "ES21", "ES22", "ES23", "ES24", "ES30",
                                "ES41", "ES42", "ES43", "ES51", "ES52", "ES53", "ES61",
                                "ES62", "FR10", "FRB0", "FRC1", "FRC2", "FRD1", "FRD2",
                                "FRE1", "FRE2", "FRF1", "FRF2", "FRF3", "FRG0", "FRH0",
                                "FRI1", "FRI2", "FRI3", "FRJ1", "FRJ2", "FRK1", "FRK2",
                                "FRL0", "ITC", "ITF", "ITG", "ITH", "ITI", "NL11", "NL12",
                                "NL13", "NL21", "NL22", "NL23", "NL31", "NL32", "NL33",
                                "NL34", "NL41", "NL42", "PT11", "PT15", "PT16", "PT17",
                                "PT18"))

# # Design weight
# weight <- ESS$dweight/mean(ESS$dweight)
# # Post-stratification weight including design weight
# weight <- ESS$pspwght/mean(ESS$pspwght)
# Analysis weight
weight <- ESS$anweight/mean(ESS$anweight)

# Gender of each respondent
gnd <- as.numeric(ESS$gndr)
# Age group of each respondent
age <- as.numeric(ESS$agecut)
# Education group of each respondent
edu <- as.numeric(ESS$educut)
# NUTS of each respondent
nuts <- as.numeric(ESS$region)
```

health - Subjective general health

```
# How is your health in general? Would you say it is...
# 1 = Very good, 2 = Good, 3 = Fair, 4 = Bad, 5 = Very bad,
# 7 = Refusal, 8 = Don't know, 9 = No answer
table(ESS$health)
```

```
##
##    1    2    3    4    5    7    8
## 3832 7761 4244 1046  154   11    5
```

```
y <- as.numeric(ESS$health)
y[y > 5] <- NA
variable <- "health"
y_labels <- c("Very good", "Good", "Fair", "Bad", "Very bad")

# survey: data frame containing the variables collected for each respondent (ESS)
# - nuts: small-area (NUTS1) from 1 to number of areas
# - gnd: 1 = Man, 2 = Woman
# - age: 1 = (14, 34]; 2 = (34, 54]; 3 = (54, 100]
# - edu: 1 = (-1, 225]; 2 = (225, 500]; 3 = (500, 1000]
# - weight: design/post-stratification/analysis weight
survey <- data.frame("y" = y, "nuts" = nuts, "gnd" = gnd,
                     "age" = age, "edu" = edu, "weight" = weight)
```

Spatial neighbourhood structure

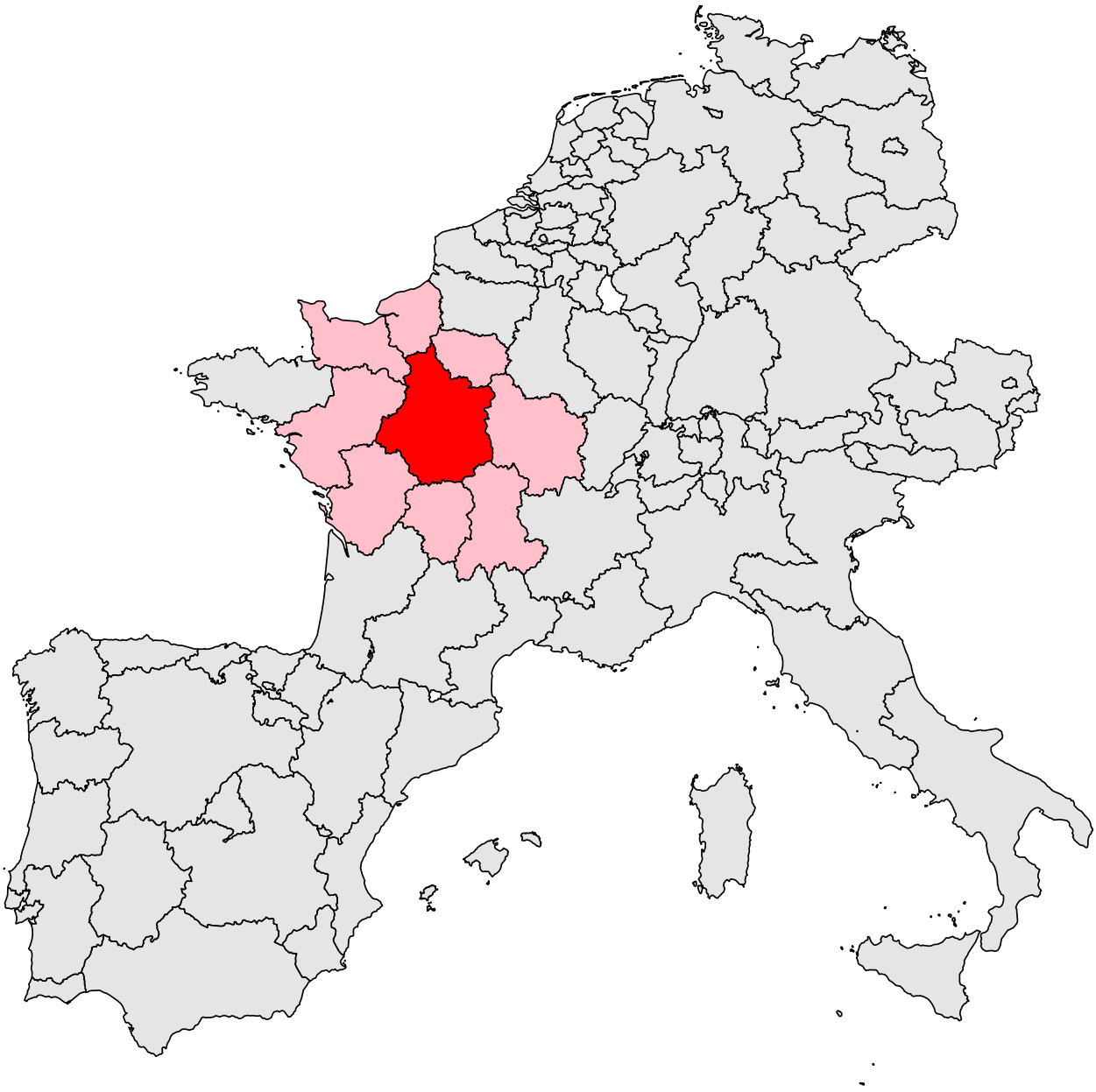
```
# Cartography of Europe
# Source: Eurostat GISCO - NUTS 2021 regions
# https://gisco-services.ec.europa.eu/distribution/v2/nuts/download/
cartography <- st_read(file.path("..", "data", "NUTS_RG_03M_2021_4326.shp"))
```

```
## Reading layer `NUTS_RG_03M_2021_4326' from data source
##   `/home/beltran_mig/Documentos/Paquete/dichotomisation/data/NUTS_RG_03M_2021_4326.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 2010 features and 8 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -63.15176 ymin: -21.38696 xmax: 55.83509 ymax: 80.83427
## Geodetic CRS:  WGS 84
```

```
cartography <- cartography[order(cartography$NUTS_ID), ]

# Western Europe
cartography <- cartography[(cartography$LEVL_CODE == 2 &
                              cartography$CNTR_CODE %in% c("AT", "BE", "CH", "ES",
                                                           "FR", "NL", "PT")) |
                             (cartography$LEVL_CODE == 1 &
                                cartography$CNTR_CODE %in% c("DE", "IT")), ]
cartography <- cartography[cartography$NUTS_ID != "ES63" &
                             cartography$NUTS_ID != "ES64" &
                             cartography$NUTS_ID != "ES70" &
                             !grepl("^FRY|^FRZ", cartography$NUTS_ID) &
                             cartography$NUTS_ID != "FRM0" &
                             cartography$NUTS_ID != "PT20" &
                             cartography$NUTS_ID != "PT30", ]
cartography <- cartography[order(cartography$NUTS_ID), ]

# Neighborhood structure by contiguity
Neigh <- poly2nb(cartography)

# Manually add adjacency between Cataluña (ES51), Comunitat Valenciana (ES52)
# and Illes Balears (ES53) to avoid treating Illes Balears as isolated.
Neigh[[which(cartography$NUTS_ID == "ES53")]] <- unique(c(Neigh[[which(cartography$NUTS_ID ==
↪  "ES53")]],
                                                          which(cartography$NUTS_ID == "ES51")))
Neigh[[which(cartography$NUTS_ID == "ES53")]] <- unique(c(Neigh[[which(cartography$NUTS_ID ==
↪  "ES53")]],
                                                          which(cartography$NUTS_ID == "ES52")))

Neigh[[which(cartography$NUTS_ID == "ES51")]] <- unique(c(Neigh[[which(cartography$NUTS_ID ==
↪  "ES51")]],
                                                          which(cartography$NUTS_ID == "ES53")))
Neigh[[which(cartography$NUTS_ID == "ES52")]] <- unique(c(Neigh[[which(cartography$NUTS_ID ==
↪  "ES52")]],
                                                          which(cartography$NUTS_ID == "ES53")))

# Adjacency matrix
W <- nb2mat(Neigh, style = "B", zero.policy = TRUE)
# D - W matrix
```

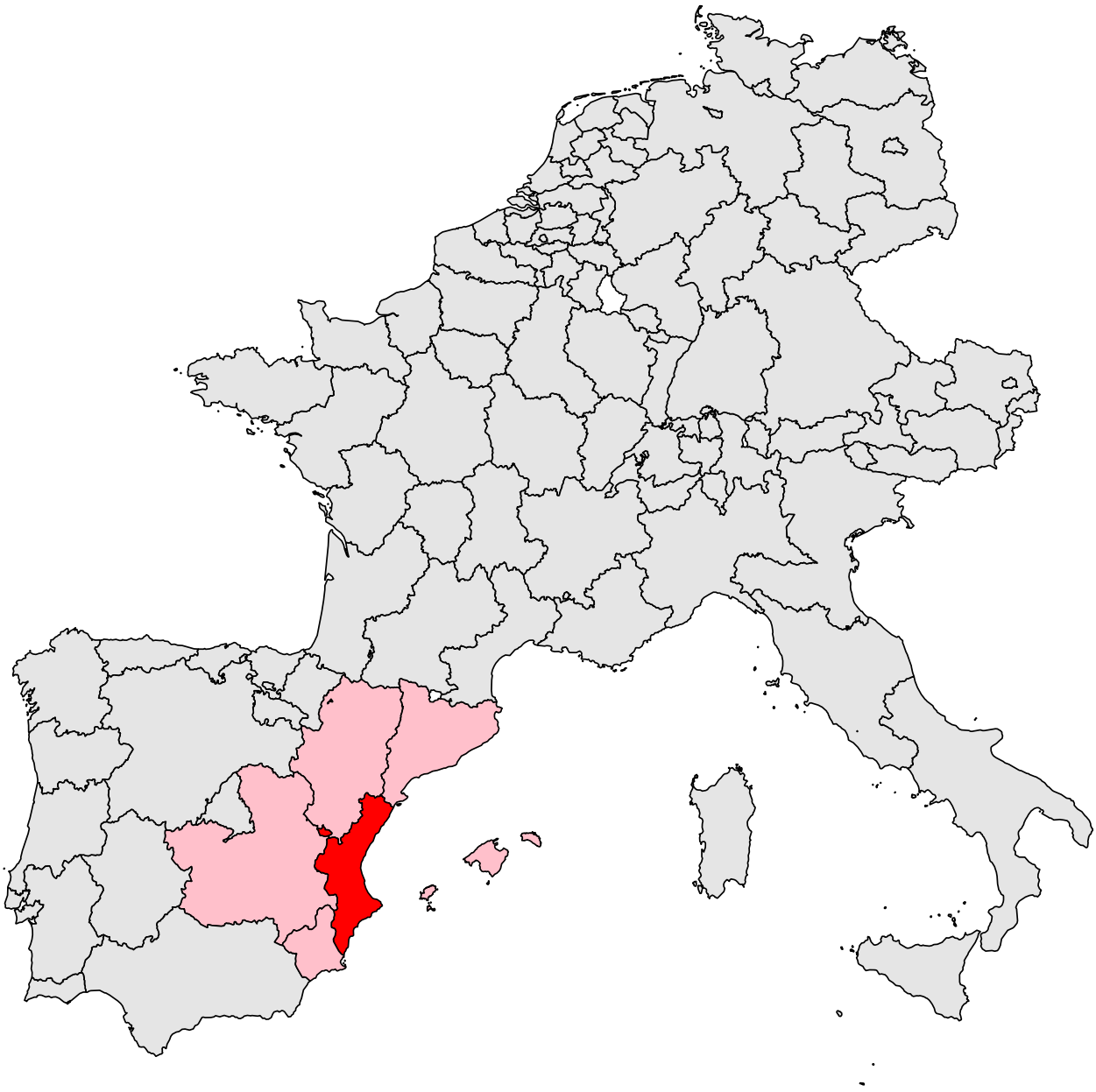
```r
Q <- diag(apply(W, 1, sum)) - W
# Number of areas
NNUTS <- nrow(W)
# Number of neighbors of each area
nadj <- apply(W, 1, sum)
# Neighbors of each area
map <- unlist(apply(W, 1, function(x) which(x != 0)))
# Sum of all the neighbor numbers of all areas
nadj.tot <- length(map)
# Cumulative sums of the number of neighbors of each area
index <- c(0, cumsum(nadj))
# All the neighborhoods j ~ i where i < j
from.to <- cbind(rep(1:NNUTS, times = nadj), map); colnames(from.to) <- c("from", "to")
from.to <- from.to[which(from.to[, 1] < from.to[, 2]), ]
NDist <- nrow(from.to)
# Eigenvalues of D - W
Lambda <- eigen(Q)$values

plot_nuts_neighbours <- function(cartography, nuts_id, nuts_col = "red",
                                 neigh_col = "pink", base_col = "grey90",
                                 border_col = "black") {

  i <- which(cartography$NUTS_ID == nuts_id)
  fill_col <- rep(base_col, nrow(cartography))
  fill_col[i] <- nuts_col
  fill_col[Neigh[[i]]] <- neigh_col

  plot(st_geometry(cartography), col = fill_col, border = border_col,
       main = paste("NUTS:", nuts_id))
}

# Plot selected regions and their neighbouring areas
plot_nuts_neighbours(cartography, nuts_id = "FRB0")
```

NUTS: FRB0



```
plot_nuts_neighbours(cartography, nuts_id = "ES52")
```

NUTS: ES52



```
cartography$NAME_LATN[cartography$NUTS_ID == "AT13"] <- "Vienna"
cartography$NAME_LATN[cartography$NUTS_ID == "BE10"] <- "Brussels-Capital Region"
cartography$NAME_LATN[cartography$NUTS_ID == "CH02"] <- "Espace Mittelland"
cartography$NAME_LATN[cartography$NUTS_ID == "DE3"]  <- "Berlin"
cartography$NAME_LATN[cartography$NUTS_ID == "ES30"] <- "Community of Madrid"
cartography$NAME_LATN[cartography$NUTS_ID == "FR10"] <- "Ile-de-France"
cartography$NAME_LATN[cartography$NUTS_ID == "ITI"]  <- "Central Italy"
cartography$NAME_LATN[cartography$NUTS_ID == "NL32"] <- "North Holland"
cartography$NAME_LATN[cartography$NUTS_ID == "PT17"] <- "Lisbon Metropolitan Area"
```

Descriptive spatial analysis

```
# Gender selection
Gender <- 1
Gender_Cat <- c("Man", "Woman")[Gender]
gender <- tolower(Gender_Cat)

ordinal_survey <- survey[survey$gnd == Gender, ]
y <- as.numeric(ordinal_survey$y)
y[y > 5] <- NA
table(y)/sum(table(y))
```

```
## y
##           1           2           3           4           5
## 0.240232443 0.469831850 0.228363007 0.053659743 0.007912957
```

```
nuts_all <- factor(ordinal_survey$nuts, levels = 1:NNUTS)
table(nuts_all)
```

```
## nuts_all
##   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
##  44 199 161  45 156 179  66  82  52  57 168  86 112  93  92  20  60  56  23  26
##  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
## 143 174  84  96 101  71  20 159 189  32  42   0  19  94  14 109 307  46  12  65
##  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
##  34  52  30  71   7  14  44  27   4  17 121  66  32  28 119  90  19 146  27 122
##  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
##  48  27  17   8  29  57  28  29  16  37  66  58  44  10  33  27  46  23  95  48
##  81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100
## 406 287 138 235 249  26  32  32  66 119  16  64 153 144  22 108  52 221  24 139
## 101 102
## 165  29
```

```
cartography$y_mean_ordinal <- as.numeric(by(y, nuts_all, mean, na.rm = TRUE))

bernoulli1_survey <- survey[survey$gnd == Gender, ]
y <- as.numeric(bernoulli1_survey$y)
y[y > 5] <- NA
y[y == 1 | y == 2] <- 0
y[y == 3 | y == 4 | y == 5] <- 1
table(y)/sum(table(y))
```

```
## y
##         0         1
## 0.7100643 0.2899357
```

```
nuts_all <- factor(bernoulli1_survey$nuts, levels = 1:NNUTS)
table(nuts_all)
```

```
## nuts_all
##   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
##  44 199 161  45 156 179  66  82  52  57 168  86 112  93  92  20  60  56  23  26
##  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
## 143 174  84  96 101  71  20 159 189  32  42   0  19  94  14 109 307  46  12  65
##  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
##  34  52  30  71   7  14  44  27   4  17 121  66  32  28 119  90  19 146  27 122
##  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
##  48  27  17   8  29  57  28  29  16  37  66  58  44  10  33  27  46  23  95  48
##  81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100
## 406 287 138 235 249  26  32  32  66 119  16  64 153 144  22 108  52 221  24 139
## 101 102
## 165  29
```

```
cartography$y_mean_bernoulli1 <- as.numeric(by(y, nuts_all, mean, na.rm = TRUE))

bernoulli2_survey <- survey[survey$gnd == Gender, ]
y <- as.numeric(bernoulli2_survey$y)
y[y > 5] <- NA
y[y == 1 | y == 2 | y == 3] <- 0
y[y == 4 | y == 5] <- 1
table(y)/sum(table(y))
```

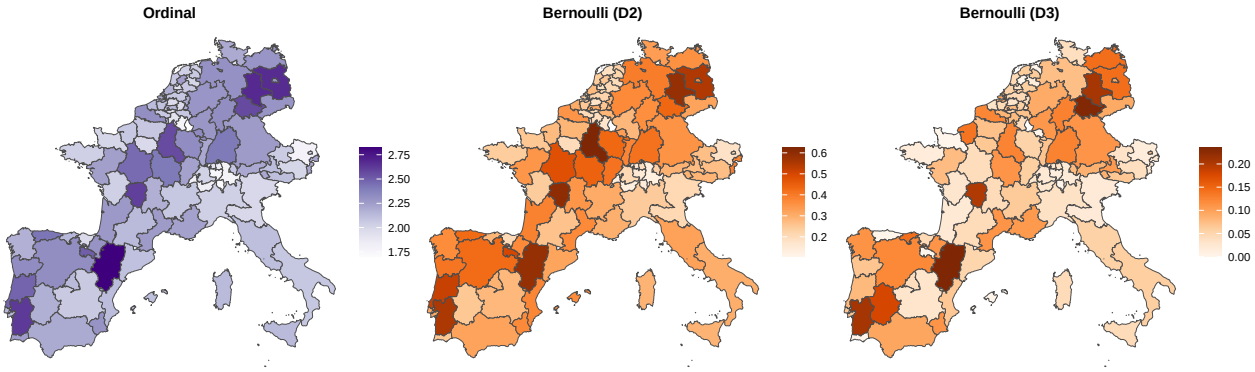
```
## y
##         0         1
## 0.9384273 0.0615727
```

```
nuts_all <- factor(bernoulli2_survey$nuts, levels = 1:NNUTS)
table(nuts_all)
```

```
## nuts_all
##   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
##  44 199 161  45 156 179  66  82  52  57 168  86 112  93  92  20  60  56  23  26
##  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
## 143 174  84  96 101  71  20 159 189  32  42   0  19  94  14 109 307  46  12  65
##  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
##  34  52  30  71   7  14  44  27   4  17 121  66  32  28 119  90  19 146  27 122
##  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
##  48  27  17   8  29  57  28  29  16  37  66  58  44  10  33  27  46  23  95  48
##  81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100
## 406 287 138 235 249  26  32  32  66 119  16  64 153 144  22 108  52 221  24 139
## 101 102
## 165  29
```

```
cartography$y_mean_bernoulli2 <- as.numeric(by(y, nuts_all, mean, na.rm = TRUE))

# Ordinal
limit <- c(min(cartography$y_mean_ordinal, na.rm = TRUE),
           max(cartography$y_mean_ordinal, na.rm = TRUE))
p_y_mean_ordinal <- ggplot(cartography) +
  geom_sf(aes(fill = y_mean_ordinal), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Purples"),
```

```
                      limits = c(limit[1], limit[2]),
                      name = NULL) + ggtitle("Ordinal") +
  theme_void() + theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12))

# Bernoulli (D2)
limit <- c(min(cartography$y_mean_bernoulli1, na.rm = TRUE),
           max(cartography$y_mean_bernoulli1, na.rm = TRUE))
p_y_mean_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = y_mean_bernoulli1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Oranges"),
                      limits = c(limit[1], limit[2]),
                      name = NULL) + ggtitle("Bernoulli (D2)") +
  theme_void() + theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12))

# Bernoulli (D3)
limit <- c(min(cartography$y_mean_bernoulli2, na.rm = TRUE),
           max(cartography$y_mean_bernoulli2, na.rm = TRUE))
p_y_mean_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = y_mean_bernoulli2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Oranges"),
                      limits = c(limit[1], limit[2]),
                      name = NULL) + ggtitle("Bernoulli (D3)") +
  theme_void() + theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12))

(p_y_mean_ordinal + p_y_mean_bernoulli1 + p_y_mean_bernoulli2)
```



Data preparation and gender filtering

```
# Gender selection
Gender <- 1
survey <- survey[survey$gnd == Gender, ]

# Number of respondents
NResp <- nrow(survey)
# Vector of zeros for the zero-trick
zero <- rep(0, NResp)

# NUTS of each respondent
nuts <- survey$nuts
# Age group of each respondent
```

```r
age <- survey$age
# Education group of each respondent
edu <- survey$edu
# Sampling weight
weight <- survey$weight

# Response variable
y <- survey$y
# Number of levels
NCats <- length(table(y))
# Vector of ones for Dirichlet prior
ones <- rep(1, NCats)

# Number of levels of each (categorical) covariate
NAge <- length(table(survey$age))
NEdu <- length(table(survey$edu))
```

Population loading

```r
# population_eu: four-dimensional array containing population counts by
# nuts, gender, age and education group.
# The population data were obtained from Eurostat:
#
↪  https://ec.europa.eu/eurostat/databrowser/view/cens_21cobe_r2/default/table?lang=en&category=cens.cens_21.cens

countries <- c("austria", "belgium", "switzerland", "germany", "spain",
               "france", "italy", "netherlands", "portugal")

population_list <- lapply(countries,
                          function(country) {
                            readRDS(file.path("..", "data", paste0("population-", country,
                            ↪  "-eu.rds")))})
population_eu <- do.call(abind, c(population_list, along = 1))

# Checking
match(dimnames(population_eu)[[1]], cartography$NUTS_ID) - 1:NNUTS
```

```
##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```r
# Gender selection
population_eu <- population_eu[, Gender, , ]
```

Ordinal model

```r
### Model code ###
```

```
modelCode <- nimbleCode(
  {
    # Likelihood
    for (Resp in 1:NResp) {
      y[Resp] ~ dcat(prlevels[Resp, 1:NCats])

      # Definition of the probabilities of each category as a function of the
      # cumulative probabilities
      prlevels[Resp, 1] <- p.gamma[Resp, 1]
      for (Cat in 2:(NCats-1)) {
        prlevels[Resp, Cat] <- p.gamma[Resp, Cat] - p.gamma[Resp, Cat-1]
      }
      prlevels[Resp, NCats] <- 1 - p.gamma[Resp, NCats-1]

      # Linear predictor
      for (Cat in 1:(NCats-1)) {
        logit(p.gamma[Resp, Cat]) <- kappa[Cat] -
          beta_age[age[Resp]] - beta_edu[edu[Resp]] -
          sd.theta * theta[nuts[Resp]]
      }
    }

    # Prior distributions

    # kappa[1:(NCats-1)] cut points
    # Monotonic transformation
    for (Cat in 1:(NCats-1)) {
      kappa[Cat] <- logit(sum(delta[1:Cat]))
    }
    # delta[1:NCats] Dirichlet prior
    delta[1:NCats] ~ ddirch(ones[1:NCats])

    # beta_age[1:NAge] age group fixed effect (corner constraint)
    beta_age[1] <- 0
    for (AgeGroup in 2:NAge) {
      beta_age[AgeGroup] ~ dflat()
    }

    # beta_edu[1:NEdu] education group fixed effect (corner constraint)
    beta_edu[1] <- 0
    for (EduGroup in 2:NEdu) {
      beta_edu[EduGroup] ~ dflat()
    }

    # theta[1:NNUTS] spatial random effect
    # LCAR distribution
    theta[1:NNUTS] ~ dcar_leroux(rho = rho,
                                 sd.theta = 1,
                                 Lambda = Lambda[1:NNUTS],
                                 from.to = from.to[1:NDist, 1:2])

    # Hyperparameters of the spatial random effect
    rho ~ dunif(0, 1)
    sd.theta ~ dhalfflat()

    # Stochastic restrictions in order to avoid confounding problems
    # Required vectors
    for (Resp in 1:NResp) {
      theta.Resp[Resp] <- theta[nuts[Resp]]
```

```
    }

    # Weighted constraint
    # Zero-mean constraint for theta.Resp[1:NResp]
    zero.theta.resp ~ dnorm(mean.thetas.resp, 10000)
    mean.thetas.resp <- mean(theta.Resp[1:NResp])

  }
)

### Data to be loaded ###

modelData <- list(y = y,
                  zero.theta.resp = 0)

modelConstants <- list(age = age, edu = edu, nuts = nuts,
                       NResp = NResp, NCats = NCats, NAge = NAge,
                       NEdu = NEdu, ones = ones, NNUTS = NNUTS,
                       NDist = NDist, Lambda = Lambda, from.to = from.to)

### Parameters to be saved ###

modelParameters <- c("kappa", "beta_age", "beta_edu",
                     "theta", "sd.theta", "rho")

### Loading functions for calling Nimble ###

source(
↪  "https://raw.githubusercontent.com/MigueBeneito/pNimble/refs/heads/main/RutinasNimble.0.2.R")
load.leroux()

# Inital values
modelInits <- local({
  constants <- modelConstants
  function() {
    library(extraDistr)
    NAge <- constants$NAge
    NEdu <- constants$NEdu; NNUTS <- constants$NNUTS
    NCats <- constants$NCats; ones <- constants$ones
    list(delta = as.numeric(rdirichlet(1, ones)),
         beta_age = c(NA, rnorm(NAge - 1)),
         beta_edu = c(NA, rnorm(NEdu - 1)),
         rho = runif(1), sd.theta = runif(1),
         theta = rnorm(NNUTS, sd = 0.1))
  }
})

# # Number of chains to run in parallel
# nchains <- 5
# # pNimble call
# salnimble <- pNimble(code = modelCode, data = modelData, constants = modelConstants,
#                      inits = modelInits, nchains = nchains, seeds = 1:nchains,
#                      niter = 2000, nburnin = 1000, thin = 5,
#                      summary = TRUE, WAIC = TRUE, monitors = modelParameters,
#                      # ntfyAccount = "MigueBeneito",
#                      HMC = TRUE, parallel = TRUE)
#
# saveRDS(salnimble, file = file.path("..", "results", paste0("ordinal-leroux-hmc-waic-", gender,
↪  ".rds")))
```

Bernoulli (D2) model

```
# Option 1 (D+): 0 = First two categories; 1 = Last three categories.
y[y > 5] <- NA
y[y == 1 | y == 2] <- 0
y[y == 3 | y == 4 | y == 5] <- 1

### Model code ###

modelCode <- nimbleCode(
  {
    # Likelihood
    for (Resp in 1:NResp) {
      y[Resp] ~ dbern(prsuccess[Resp])

      # Linear predictor
      logit(prsuccess[Resp]) <- beta_0 +
        beta_age[age[Resp]] + beta_edu[edu[Resp]] +
        sd.theta * theta[nuts[Resp]]
    }

    # Prior distributions

    # beta_0 intercept
    beta_0 ~ dflat()

    # beta_age[1:NAge] age group fixed effect (corner constraint)
    beta_age[1] <- 0
    for (AgeGroup in 2:NAge) {
      beta_age[AgeGroup] ~ dflat()
    }

    # beta_edu[1:NEdu] education group fixed effect (corner constraint)
    beta_edu[1] <- 0
    for (EduGroup in 2:NEdu) {
      beta_edu[EduGroup] ~ dflat()
    }

    # theta[1:NNUTS] spatial random effect
    # LCAR distribution
    theta[1:NNUTS] ~ dcar_leroux(rho = rho,
                                 sd.theta = 1,
                                 Lambda = Lambda[1:NNUTS],
                                 from.to = from.to[1:NDist, 1:2])

    # Hyperparameters of the spatial random effect
    rho ~ dunif(0, 1)
    sd.theta ~ dhalfflat()

    # Stochastic restrictions in order to avoid confounding problems
    # Required vectors
    for (Resp in 1:NResp) {
      theta.Resp[Resp] <- theta[nuts[Resp]]
    }

    # Weighted constraint
    # Zero-mean constraint for theta.Resp[1:NResp]
    zero.theta.resp ~ dnorm(mean.thetas.resp, 10000)
```

```
    mean.thetas.resp <- mean(theta.Resp[1:NResp])

  }
)

### Data to be loaded ###

modelData <- list(y = y,
                  zero.theta.resp = 0)

modelConstants <- list(age = age, edu = edu, nuts = nuts,
                       NResp = NResp, NAge = NAge,
                       NEdu = NEdu, NNUTS = NNUTS,
                       NDist = NDist, Lambda = Lambda, from.to = from.to)

### Parameters to be saved ###

modelParameters <- c("beta_0", "beta_age", "beta_edu",
                     "theta", "sd.theta", "rho")

### Loading functions for calling Nimble ###

source(
↪  "https://raw.githubusercontent.com/MigueBeneito/pNimble/refs/heads/main/RutinasNimble.0.2.R")
load.leroux()

# Inital values
modelInits <- local({
  constants <- modelConstants
  function() {
    NAge <- constants$NAge
    NEdu <- constants$NEdu; NNUTS <- constants$NNUTS
    list(beta_0 = rnorm(1),
         beta_age = c(NA, rnorm(NAge - 1)),
         beta_edu = c(NA, rnorm(NEdu - 1)),
         rho = runif(1), sd.theta = runif(1),
         theta = rnorm(NNUTS, sd = 0.1))
  }
})

# # Number of chains to run in parallel
# nchains <- 5
# # pNimble call
# salnimble <- pNimble(code = modelCode, data = modelData, constants = modelConstants,
#                      inits = modelInits, nchains = nchains, seeds = 1:nchains,
#                      niter = 2000, nburnin = 1000, thin = 5,
#                      summary = TRUE, WAIC = TRUE, monitors = modelParameters,
#                      # ntfyAccount = "MigueBeneito",
#                      HMC = TRUE, parallel = TRUE)
#
# saveRDS(salnimble, file = file.path("..", "results", paste0("bernoulli-leroux-hmc-waic-",
↪  gender, "-d2.rds")))
```

Bernoulli (D3) model

```
# Option 2 (D-): 0 = First three categories; 1 = Last two categories.
y[y > 5] <- NA
y[y == 1 | y == 2 | y == 3] <- 0
y[y == 4 | y == 5] <- 1

### Model code ###

modelCode <- nimbleCode(
  {
    # Likelihood
    for (Resp in 1:NResp) {
      y[Resp] ~ dbern(prsuccess[Resp])

      # Linear predictor
      logit(prsuccess[Resp]) <- beta_0 +
        beta_age[age[Resp]] + beta_edu[edu[Resp]] +
        sd.theta * theta[nuts[Resp]]
    }

    # Prior distributions

    # beta_0 intercept
    beta_0 ~ dflat()

    # beta_age[1:NAge] age group fixed effect (corner constraint)
    beta_age[1] <- 0
    for (AgeGroup in 2:NAge) {
      beta_age[AgeGroup] ~ dflat()
    }

    # beta_edu[1:NEdu] education group fixed effect (corner constraint)
    beta_edu[1] <- 0
    for (EduGroup in 2:NEdu) {
      beta_edu[EduGroup] ~ dflat()
    }

    # theta[1:NNUTS] spatial random effect
    # LCAR distribution
    theta[1:NNUTS] ~ dcar_leroux(rho = rho,
                                 sd.theta = 1,
                                 Lambda = Lambda[1:NNUTS],
                                 from.to = from.to[1:NDist, 1:2])

    # Hyperparameters of the spatial random effect
    rho ~ dunif(0, 1)
    sd.theta ~ dhalfflat()

    # Stochastic restrictions in order to avoid confounding problems
    # Required vectors
    for (Resp in 1:NResp) {
      theta.Resp[Resp] <- theta[nuts[Resp]]
    }

    # Weighted constraint
    # Zero-mean constraint for theta.Resp[1:NResp]
    zero.theta.resp ~ dnorm(mean.thetas.resp, 10000)
```

```
    mean.thetas.resp <- mean(theta.Resp[1:NResp])

  }
)

### Data to be loaded ###

modelData <- list(y = y,
                  zero.theta.resp = 0)

modelConstants <- list(age = age, edu = edu, nuts = nuts,
                       NResp = NResp, NAge = NAge,
                       NEdu = NEdu, NNUTS = NNUTS,
                       NDist = NDist, Lambda = Lambda, from.to = from.to)

### Parameters to be saved ###

modelParameters <- c("beta_0", "beta_age", "beta_edu",
                     "theta", "sd.theta", "rho")

### Loading functions for calling Nimble ###

source(
↪  "https://raw.githubusercontent.com/MigueBeneito/pNimble/refs/heads/main/RutinasNimble.0.2.R")
load.leroux()

# Inital values
modelInits <- local({
  constants <- modelConstants
  function() {
    NAge <- constants$NAge
    NEdu <- constants$NEdu; NNUTS <- constants$NNUTS
    list(beta_0 = rnorm(1),
         beta_age = c(NA, rnorm(NAge - 1)),
         beta_edu = c(NA, rnorm(NEdu - 1)),
         rho = runif(1), sd.theta = runif(1),
         theta = rnorm(NNUTS, sd = 0.1))
  }
})

# # Number of chains to run in parallel
# nchains <- 5
# # pNimble call
# salnimble <- pNimble(code = modelCode, data = modelData, constants = modelConstants,
#                      inits = modelInits, nchains = nchains, seeds = 1:nchains,
#                      niter = 2000, nburnin = 1000, thin = 5,
#                      summary = TRUE, WAIC = TRUE, monitors = modelParameters,
#                      # ntfyAccount = "MigueBeneito",
#                      HMC = TRUE, parallel = TRUE)
#
# saveRDS(salnimble, file = file.path("..", "results", paste0("bernoulli-leroux-hmc-waic-",
↪  gender, "-d3.rds")))
```

Loading posterior samples

```
# Ordinal regression
ordinal_results <- readRDS(file = file.path("..", "results", paste0("ordinal-leroux-hmc-waic-",
↪  gender, ".rds")))
# Logistic regression (Option 1): 0 = First two categories; 1 = Last three categories.
bernoulli_results1 <- readRDS(file = file.path("..", "results",
↪  paste0("bernoulli-leroux-hmc-waic-", gender, "-d2.rds")))
# Logistic regression (Option 2): 0 = First three categories; 1 = Last two categories.
bernoulli_results2 <- readRDS(file = file.path("..", "results",
↪  paste0("bernoulli-leroux-hmc-waic-", gender, "-d3.rds")))
```

Convert NIMBLE output to WinBUGS-style format

```
nchains <- 5
nsims <- nchains * nrow(ordinal_results$samples[[1]])

# NimToWin:
# - transforms NIMBLE output to WinBUGS sims.list output format

NimToWin <- function(salnimble) {

  kappa <- matrix(nrow = nsims, ncol = NCats - 1)
  beta_age <- matrix(nrow = nsims, ncol = NAge)
  beta_edu <- matrix(nrow = nsims, ncol = NEdu)
  theta <- matrix(nrow = nsims, ncol = NNUTS)
  sd.theta <- numeric(length = nsims)
  rho <- numeric(length = nsims)

  for (Cat in 1:(NCats - 1)) {
    kappa[, Cat] <- c(salnimble[[1]][,  paste0("kappa[", Cat, "]")],
                      salnimble[[2]][,  paste0("kappa[", Cat, "]")],
                      salnimble[[3]][,  paste0("kappa[", Cat, "]")],
                      salnimble[[4]][,  paste0("kappa[", Cat, "]")],
                      salnimble[[5]][,  paste0("kappa[", Cat, "]")])
  }

  for (AgeGroup in 1:NAge) {
    beta_age[, AgeGroup] <- c(salnimble[[1]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[2]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[3]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[4]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[5]][, paste0("beta_age[", AgeGroup, "]")])
  }

  for (EduGroup in 1:NEdu) {
    beta_edu[, EduGroup] <- c(salnimble[[1]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[2]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[3]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[4]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[5]][, paste0("beta_edu[", EduGroup, "]")])
  }
```

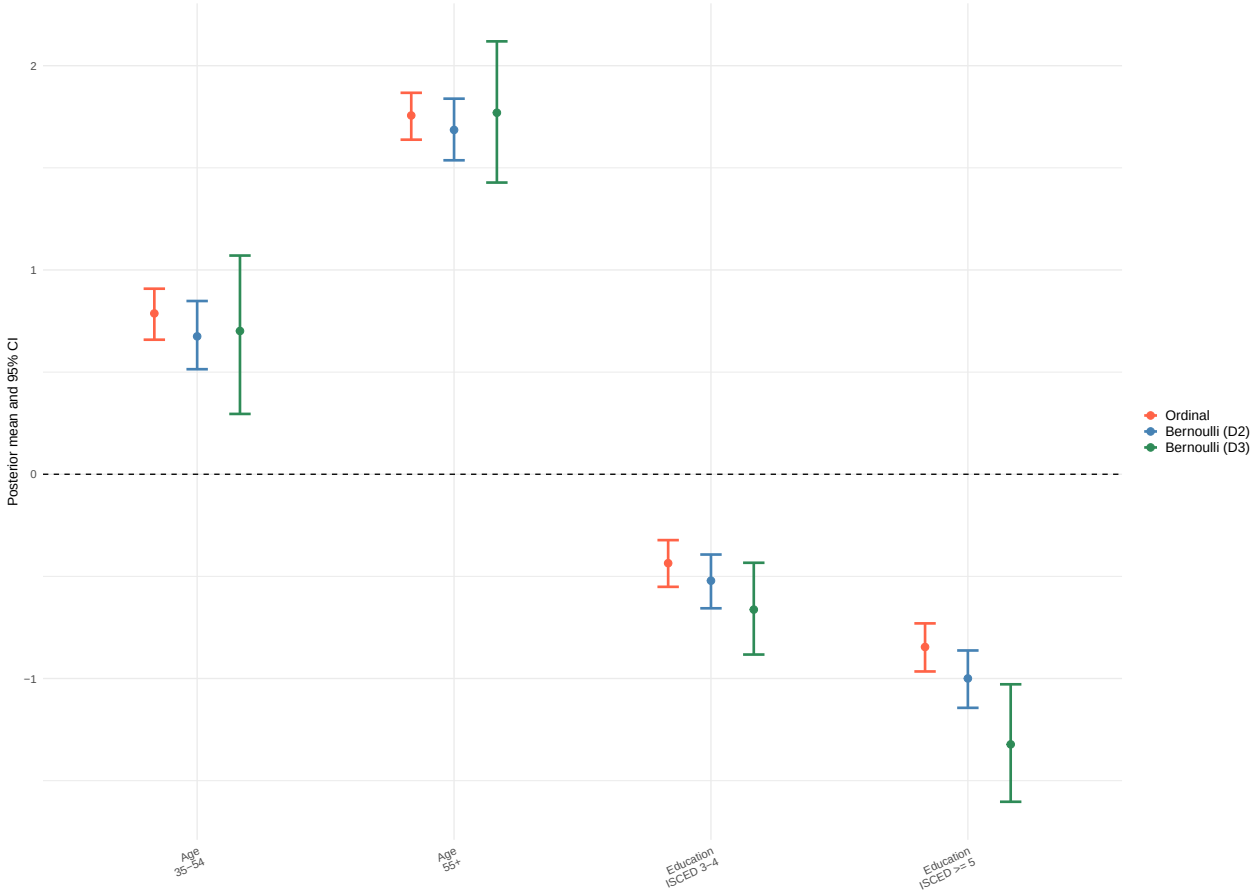
```
  for (NUTS in 1:NNUTS) {
    theta[, NUTS] <- c(salnimble[[1]][, paste0("theta[", NUTS, "]")],
                       salnimble[[2]][, paste0("theta[", NUTS, "]")],
                       salnimble[[3]][, paste0("theta[", NUTS, "]")],
                       salnimble[[4]][, paste0("theta[", NUTS, "]")],
                       salnimble[[5]][, paste0("theta[", NUTS, "]")])
  }

  sd.theta <- c(salnimble[[1]][, "sd.theta"], salnimble[[2]][, "sd.theta"],
                salnimble[[3]][, "sd.theta"], salnimble[[4]][, "sd.theta"],
                salnimble[[5]][, "sd.theta"])

  rho <- c(salnimble[[1]][, "rho"], salnimble[[2]][, "rho"],
           salnimble[[3]][, "rho"], salnimble[[4]][, "rho"],
           salnimble[[5]][, "rho"])

  summary <- MCMCsummary(object = salnimble, round = 4)
  sims.list <- list("kappa" = kappa, "beta_age" = beta_age, "beta_edu" = beta_edu,
                    "theta" = theta, "sd.theta" = sd.theta, "rho" = rho)

  salwinbugs <- list("summary" = summary, "sims.list" = sims.list,
                     "nchains" = nchains, "nsims" = nsims)

  return(salwinbugs)
}

ordinal_salwinbugs <- NimToWin(salnimble = ordinal_results$samples)

NimToWin <- function(salnimble) {

  beta_0 <- numeric(length = nsims)
  beta_age <- matrix(nrow = nsims, ncol = NAge)
  beta_edu <- matrix(nrow = nsims, ncol = NEdu)
  theta <- matrix(nrow = nsims, ncol = NNUTS)
  sd.theta <- numeric(length = nsims)
  rho <- numeric(length = nsims)

  beta_0 <- c(salnimble[[1]][, "beta_0"], salnimble[[2]][, "beta_0"],
              salnimble[[3]][, "beta_0"], salnimble[[4]][, "beta_0"],
              salnimble[[5]][, "beta_0"])

  for (AgeGroup in 1:NAge) {
    beta_age[, AgeGroup] <- c(salnimble[[1]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[2]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[3]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[4]][, paste0("beta_age[", AgeGroup, "]")],
                              salnimble[[5]][, paste0("beta_age[", AgeGroup, "]")])
  }

  for (EduGroup in 1:NEdu) {
    beta_edu[, EduGroup] <- c(salnimble[[1]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[2]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[3]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[4]][, paste0("beta_edu[", EduGroup, "]")],
                              salnimble[[5]][, paste0("beta_edu[", EduGroup, "]")])
  }

  for (NUTS in 1:NNUTS) {
    theta[, NUTS] <- c(salnimble[[1]][, paste0("theta[", NUTS, "]")],
```

```
                      salnimble[[2]][, paste0("theta[", NUTS, "]")],
                      salnimble[[3]][, paste0("theta[", NUTS, "]")],
                      salnimble[[4]][, paste0("theta[", NUTS, "]")],
                      salnimble[[5]][, paste0("theta[", NUTS, "]")])
  }

  sd.theta <- c(salnimble[[1]][, "sd.theta"], salnimble[[2]][, "sd.theta"],
                salnimble[[3]][, "sd.theta"], salnimble[[4]][, "sd.theta"],
                salnimble[[5]][, "sd.theta"])

  rho <- c(salnimble[[1]][, "rho"], salnimble[[2]][, "rho"],
           salnimble[[3]][, "rho"], salnimble[[4]][, "rho"],
           salnimble[[5]][, "rho"])

  summary <- MCMCsummary(object = salnimble, round = 4)
  sims.list <- list("beta_0" = beta_0, "beta_age" = beta_age, "beta_edu" = beta_edu,
                    "theta" = theta, "sd.theta" = sd.theta, "rho" = rho)

  salwinbugs <- list("summary" = summary, "sims.list" = sims.list,
                     "nchains" = nchains, "nsims" = nsims)

  return(salwinbugs)
}

bernoulli1_salwinbugs <- NimToWin(salnimble = bernoulli_results1$samples)
bernoulli2_salwinbugs <- NimToWin(salnimble = bernoulli_results2$samples)
```

Covariate effects

```
df_plot_ord <- data.frame("param" = rownames(ordinal_results$summary)[1:6],
                          "mean" = ordinal_results$summary[1:6, 1],
                          "lower" = ordinal_results$summary[1:6, 3],
                          "upper" = ordinal_results$summary[1:6, 5],
                          "model" = "Ordinal")
df_plot_bern1 <- data.frame("param" = rownames(bernoulli_results1$summary)[2:7],
                            "mean" = bernoulli_results1$summary[2:7, 1],
                            "lower" = bernoulli_results1$summary[2:7, 3],
                            "upper" = bernoulli_results1$summary[2:7, 5],
                            "model" = "Bernoulli (D2)")
df_plot_bern2 <- data.frame("param" = rownames(bernoulli_results2$summary)[2:7],
                            "mean" = bernoulli_results2$summary[2:7, 1],
                            "lower" = bernoulli_results2$summary[2:7, 3],
                            "upper" = bernoulli_results2$summary[2:7, 5],
                            "model" = "Bernoulli (D3)")
# df_plot <- rbind(df_plot_ord, df_plot_bern1, df_plot_bern2)
# df_plot$param <- factor(df_plot$param, levels = df_plot$param[1:6])
# df_plot$model <- factor(df_plot$model, levels = c("Ordinal", "Bernoulli (D2)", "Bernoulli
↪  (D3)"))

df_plot <- rbind(df_plot_ord, df_plot_bern1, df_plot_bern2)
df_plot <- subset(df_plot, !(param %in% c("beta_age[1]", "beta_edu[1]")))
df_plot$param <- factor(df_plot$param,
                        levels = c("beta_age[2]", "beta_age[3]", "beta_edu[2]", "beta_edu[3]"),
                        labels = c("Age\n35-54", "Age\n55+", "Education\nISCED 3-4",
                        ↪  "Education\nISCED >= 5"))
```

```r
df_plot$model <- factor(df_plot$model, levels = c("Ordinal", "Bernoulli (D2)", "Bernoulli (D3)"))

p_fixed <- ggplot(df_plot, aes(x = param, y = mean, color = model)) +
  geom_point(position = position_dodge(width = 0.5), size = 2.5) +
  geom_errorbar(aes(ymin = lower, ymax = upper),
                position = position_dodge(width = 0.5), width = 0.25, linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_color_manual(values = c("Ordinal" = "tomato",
                                "Bernoulli (D2)" = "steelblue",
                                "Bernoulli (D3)" = "seagreen"),
                     labels = c("Ordinal", "Bernoulli (D2)", "Bernoulli (D3)")) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1),
        legend.text = element_text(size = 11),
        legend.key.size = unit(0.45, "cm")) +
  labs(x = NULL, y = "Posterior mean and 95% CI", color = "")

p_fixed
```



Spatial effect

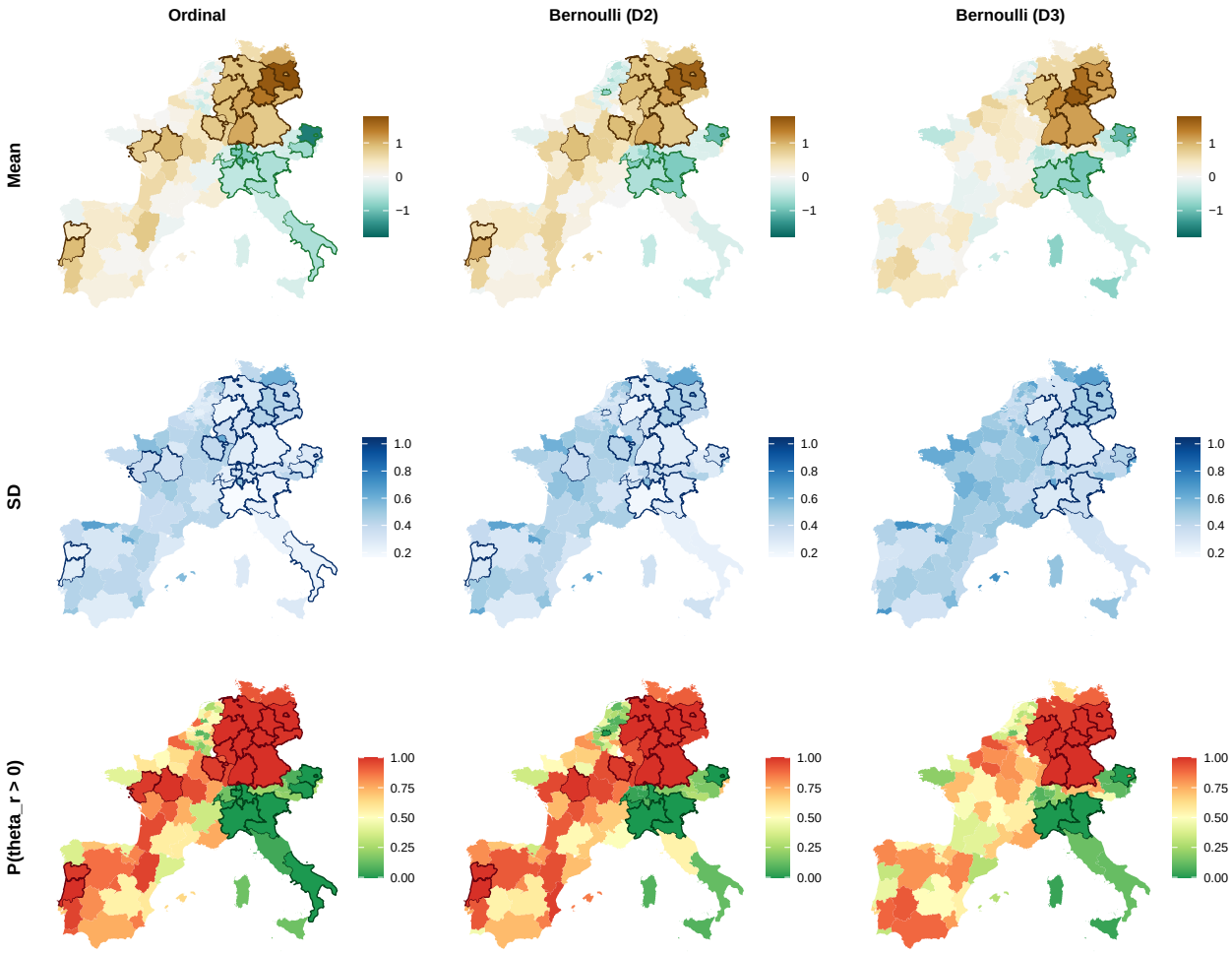
```r
# Posterior mean of theta's
cartography$thetamean_ordinal <-
↪  ordinal_results$summary[startsWith(rownames(ordinal_results$summary), "theta"), 1]
```

```
cartography$thetamean_bernoulli1 <-
↪  bernoulli_results1$summary[startsWith(rownames(bernoulli_results1$summary), "theta"), 1]
cartography$thetamean_bernoulli2 <-
↪  bernoulli_results2$summary[startsWith(rownames(bernoulli_results2$summary), "theta"), 1]

# 95% credible intervals of theta's
cartography$thetalow_ordinal <-
↪  ordinal_results$summary[startsWith(rownames(ordinal_results$summary), "theta"), 3]
cartography$thetaup_ordinal <-
↪  ordinal_results$summary[startsWith(rownames(ordinal_results$summary), "theta"), 5]
cartography$thetalow_bernoulli1 <-
↪  bernoulli_results1$summary[startsWith(rownames(bernoulli_results1$summary), "theta"), 3]
cartography$thetaup_bernoulli1 <-
↪  bernoulli_results1$summary[startsWith(rownames(bernoulli_results1$summary), "theta"), 5]
cartography$thetalow_bernoulli2 <-
↪  bernoulli_results2$summary[startsWith(rownames(bernoulli_results2$summary), "theta"), 3]
cartography$thetaup_bernoulli2 <-
↪  bernoulli_results2$summary[startsWith(rownames(bernoulli_results2$summary), "theta"), 5]

# Posterior sd of theta's
cartography$thetasd_ordinal <-
↪  ordinal_results$summary[startsWith(rownames(ordinal_results$summary), "theta"), 2]
cartography$thetasd_bernoulli1 <-
↪  bernoulli_results1$summary[startsWith(rownames(bernoulli_results1$summary), "theta"), 2]
cartography$thetasd_bernoulli2 <-
↪  bernoulli_results2$summary[startsWith(rownames(bernoulli_results2$summary), "theta"), 2]

# Checking when theta is greater than zero
ordinal_stepsim <- 1 * (ordinal_salwinbugs$sims.list$theta > 0)
bernoulli1_stepsim <- 1 * (bernoulli1_salwinbugs$sims.list$theta > 0)
bernoulli2_stepsim <- 1 * (bernoulli2_salwinbugs$sims.list$theta > 0)

# Posterior probabilities of theta's
cartography$probmean_ordinal <- apply(ordinal_stepsim, 2, mean)
cartography$probmean_bernoulli1 <- apply(bernoulli1_stepsim, 2, mean)
cartography$probmean_bernoulli2 <- apply(bernoulli2_stepsim, 2, mean)

# 95% credible intervals of posterior probabilities
cartography$problow_ordinal <- apply(ordinal_stepsim, 2, quantile, probs = 0.025)
cartography$probup_ordinal <- apply(ordinal_stepsim, 2, quantile, probs = 0.975)
cartography$problow_bernoulli1 <- apply(bernoulli1_stepsim, 2, quantile, probs = 0.025)
cartography$probup_bernoulli1 <- apply(bernoulli1_stepsim, 2, quantile, probs = 0.975)
cartography$problow_bernoulli2 <- apply(bernoulli2_stepsim, 2, quantile, probs = 0.025)
cartography$probup_bernoulli2 <- apply(bernoulli2_stepsim, 2, quantile, probs = 0.975)

# Borders for mean and sd rows: theta interval does not include zero
cartography$border_mean_ordinal <- NA
cartography$border_mean_ordinal[cartography$thetalow_ordinal > 0] <- "#543005"
cartography$border_mean_ordinal[cartography$thetaup_ordinal < 0] <- "#1B7837"

cartography$border_mean_bernoulli1 <- NA
cartography$border_mean_bernoulli1[cartography$thetalow_bernoulli1 > 0] <- "#543005"
cartography$border_mean_bernoulli1[cartography$thetaup_bernoulli1 < 0] <- "#1B7837"

cartography$border_mean_bernoulli2 <- NA
cartography$border_mean_bernoulli2[cartography$thetalow_bernoulli2 > 0] <- "#543005"
cartography$border_mean_bernoulli2[cartography$thetaup_bernoulli2 < 0] <- "#1B7837"

cartography$border_sd_ordinal <- NA
```

```r
cartography$border_sd_ordinal[cartography$thetalow_ordinal > 0 | cartography$thetaup_ordinal < 0]
↪  <- "#08306B"

cartography$border_sd_bernoulli1 <- NA
cartography$border_sd_bernoulli1[cartography$thetalow_bernoulli1 > 0 |
↪  cartography$thetaup_bernoulli1 < 0] <- "#08306B"

cartography$border_sd_bernoulli2 <- NA
cartography$border_sd_bernoulli2[cartography$thetalow_bernoulli2 > 0 |
↪  cartography$thetaup_bernoulli2 < 0] <- "#08306B"

# Borders for significance row: probability interval does not include 0.5
cartography$border_prob_ordinal <- NA
cartography$border_prob_ordinal[cartography$problow_ordinal > 0.5] <- "#67000D"
cartography$border_prob_ordinal[cartography$probup_ordinal < 0.5] <- "#00441B"

cartography$border_prob_bernoulli1 <- NA
cartography$border_prob_bernoulli1[cartography$problow_bernoulli1 > 0.5] <- "#67000D"
cartography$border_prob_bernoulli1[cartography$probup_bernoulli1 < 0.5] <- "#00441B"

cartography$border_prob_bernoulli2 <- NA
cartography$border_prob_bernoulli2[cartography$problow_bernoulli2 > 0.5] <- "#67000D"
cartography$border_prob_bernoulli2[cartography$probup_bernoulli2 < 0.5] <- "#00441B"

# Mean
limit <- max(abs(c(cartography$thetamean_ordinal,
                   cartography$thetamean_bernoulli1,
                   cartography$thetamean_bernoulli2)), na.rm = TRUE)

p_thetamean_ordinal <- ggplot(cartography) +
  geom_sf(aes(fill = thetamean_ordinal, color = border_mean_ordinal), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "BrBG")[9:1],
                       limits = c(-limit, limit),
                       values = rescale(c(-limit, 0, limit)),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_thetamean_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = thetamean_bernoulli1, color = border_mean_bernoulli1), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "BrBG")[9:1],
                       limits = c(-limit, limit),
                       values = rescale(c(-limit, 0, limit)),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_thetamean_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = thetamean_bernoulli2, color = border_mean_bernoulli2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "BrBG")[9:1],
                       limits = c(-limit, limit),
                       values = rescale(c(-limit, 0, limit)),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

# Sd
limit <- c(min(c(cartography$thetasd_ordinal,
                 cartography$thetasd_bernoulli1,
```

```
                cartography$thetasd_bernoulli2), na.rm = TRUE),
          max(c(cartography$thetasd_ordinal,
                cartography$thetasd_bernoulli1,
                cartography$thetasd_bernoulli2), na.rm = TRUE))

p_thetasd_ordinal <- ggplot(cartography) +
  geom_sf(aes(fill = thetasd_ordinal, color = border_sd_ordinal), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = c(limit[1], limit[2]),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_thetasd_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = thetasd_bernoulli1, color = border_sd_bernoulli1), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = c(limit[1], limit[2]),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_thetasd_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = thetasd_bernoulli2, color = border_sd_bernoulli2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = c(limit[1], limit[2]),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

# Significance
p_probmean_ordinal <- ggplot(cartography) +
  geom_sf(aes(fill = probmean_ordinal, color = border_prob_ordinal), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlGn")[9:1],
                       limits = c(0, 1),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_probmean_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = probmean_bernoulli1, color = border_prob_bernoulli1), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlGn")[9:1],
                       limits = c(0, 1),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_probmean_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = probmean_bernoulli2, color = border_prob_bernoulli2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlGn")[9:1],
                       limits = c(0, 1),
                       name = NULL) +
  scale_color_identity(na.value = NA) +
  theme_void()

p_thetamean_ordinal <- p_thetamean_ordinal + ggtitle("Ordinal")
p_thetamean_bernoulli1 <- p_thetamean_bernoulli1 + ggtitle("Bernoulli (D2)")
p_thetamean_bernoulli2 <- p_thetamean_bernoulli2 + ggtitle("Bernoulli (D3)")

p_thetamean_ordinal <- p_thetamean_ordinal + labs(tag = "Mean")
```

```r
p_thetasd_ordinal <- p_thetasd_ordinal + labs(tag = "SD")
p_probmean_ordinal <- p_probmean_ordinal + labs(tag = "P(theta_r > 0)")

tema_mapas <- theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
                    plot.tag = element_text(face = "bold", size = 13, angle = 90),
                    plot.tag.position = c(-0.08, 0.5),
                    plot.margin = margin(5.5, 5.5, 5.5, 20))

final_plot <- wrap_plots(p_thetamean_ordinal + tema_mapas,
                         p_thetamean_bernoulli1 + tema_mapas,
                         p_thetamean_bernoulli2 + tema_mapas,
                         p_thetasd_ordinal + tema_mapas,
                         p_thetasd_bernoulli1 + tema_mapas,
                         p_thetasd_bernoulli2 + tema_mapas,
                         p_probmean_ordinal + tema_mapas,
                         p_probmean_bernoulli1 + tema_mapas,
                         p_probmean_bernoulli2 + tema_mapas, ncol = 3)

final_plot
```



Poststratified estimates

```
# poststratify:
# - (1) computes the nsims simulated probabilities for each age, education and nuts
# - (2) post-stratifies these probabilities for each nuts and category

poststratify <- function(salwinbugs) {

  p.gamma <- array(dim = c(nsims, NNUTS, NAge, NEdu, NCats - 1))
  prlevels <- array(dim = c(nsims, NNUTS, NAge, NEdu, NCats))
  prlevels_post <- array(dim = c(nsims, NNUTS, NCats))
  population <- population_eu

  # Probabilities for each age, education and nuts
  for (sim in 1:nsims) {
    for (AgeGroup in 1:NAge) {
      for (EduGroup in 1:NEdu) {
        for (NUTS in 1:NNUTS) {
          for (Cat in 1:(NCats - 1)) {
            p.gamma[sim, NUTS, AgeGroup, EduGroup, Cat] <-
              ilogit(salwinbugs$sims.list$kappa[sim, Cat] -
                       salwinbugs$sims.list$beta_age[sim, AgeGroup] -
                       salwinbugs$sims.list$beta_edu[sim, EduGroup] -
                       salwinbugs$sims.list$sd.theta[sim] * salwinbugs$sims.list$theta[sim,
↪  NUTS])
          }

          prlevels[sim, NUTS, AgeGroup, EduGroup, 1] <- p.gamma[sim, NUTS, AgeGroup, EduGroup, 1]
          prlevels[sim, NUTS, AgeGroup, EduGroup, NCats] <- 1 - p.gamma[sim, NUTS, AgeGroup,
↪  EduGroup, NCats - 1]

          for (Cat in 2:(NCats - 1)) {
            prlevels[sim, NUTS, AgeGroup, EduGroup, Cat] <-
              p.gamma[sim, NUTS, AgeGroup, EduGroup, Cat] - p.gamma[sim, NUTS, AgeGroup,
↪  EduGroup, Cat-1]
          }
        }
      }
    }

    # Post-stratification
    for (NUTS in 1:NNUTS) {
      for (Cat in 1:NCats) {
        prlevels_post[sim, NUTS, Cat] <- sum(population[NUTS, , ])^(-1) * sum(prlevels[sim, NUTS,
↪  , , Cat] * population[NUTS, , ])
      }
    }
    if (sim %in% c(1, seq(nsims/nchains, nsims, nsims/nchains))) {
      cat(sim, "of", nsims, "simulations", "\n")
    } else {}
  }

  return(prlevels_post)
}

ordinal_post <- poststratify(salwinbugs = ordinal_salwinbugs)
```

```
## 1 of 1000 simulations
## 200 of 1000 simulations
## 400 of 1000 simulations
## 600 of 1000 simulations
## 800 of 1000 simulations
## 1000 of 1000 simulations
```

```
poststratify <- function(salwinbugs) {

  prsuccess <- array(dim = c(nsims, NNUTS, NAge, NEdu))
  prsuccess_post <- matrix(nrow = nsims, ncol = NNUTS)
  population <- population_eu

  # Probabilities for each age, education and nuts
  for (sim in 1:nsims) {
    for (AgeGroup in 1:NAge) {
      for (EduGroup in 1:NEdu) {
        for (NUTS in 1:NNUTS) {
          prsuccess[sim, NUTS, AgeGroup, EduGroup] <-
            ilogit(salwinbugs$sims.list$beta_0[sim] +
                     salwinbugs$sims.list$beta_age[sim, AgeGroup] +
                     salwinbugs$sims.list$beta_edu[sim, EduGroup] +
                     salwinbugs$sims.list$sd.theta[sim] * salwinbugs$sims.list$theta[sim, NUTS])
        }
      }
    }

    # Post-stratification
    for (NUTS in 1:NNUTS) {
      prsuccess_post[sim, NUTS] <- sum(population[NUTS, , ])^(-1) * sum(prsuccess[sim, NUTS, , ]
↪  * population[NUTS, , ])
    }
    if (sim %in% c(1, seq(nsims/nchains, nsims, nsims/nchains))) {
      cat(sim, "of", nsims, "simulations", "\n")
    } else {}
  }

  return(prsuccess_post)
}

bernoulli1_post <- poststratify(salwinbugs = bernoulli1_salwinbugs)
```

```
## 1 of 1000 simulations
## 200 of 1000 simulations
## 400 of 1000 simulations
## 600 of 1000 simulations
## 800 of 1000 simulations
## 1000 of 1000 simulations
```

```
bernoulli2_post <- poststratify(salwinbugs = bernoulli2_salwinbugs)
```

```
## 1 of 1000 simulations
## 200 of 1000 simulations
## 400 of 1000 simulations
## 600 of 1000 simulations
```

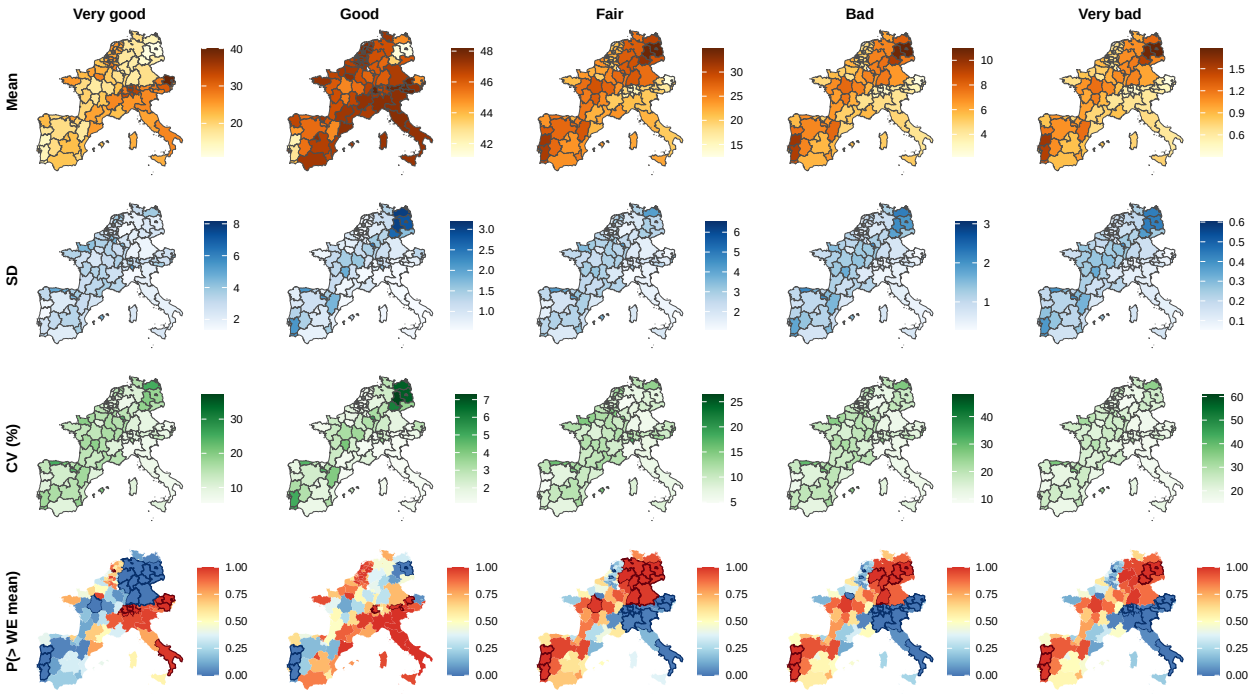
```
## 800 of 1000 simulations
## 1000 of 1000 simulations
```

```r
# Post-stratified posterior means and sd's of the percentages for each nuts and category

### Ordinal model ###

# Mean
cartography$percentage_mean1 <- apply(ordinal_post, 2:3, mean)[, 1] * 100
p_percentage_mean1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  theme_void()
cartography$percentage_mean2 <- apply(ordinal_post, 2:3, mean)[, 2] * 100
p_percentage_mean2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  theme_void()
cartography$percentage_mean3 <- apply(ordinal_post, 2:3, mean)[, 3] * 100
p_percentage_mean3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  theme_void()
cartography$percentage_mean4 <- apply(ordinal_post, 2:3, mean)[, 4] * 100
p_percentage_mean4 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean4), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  theme_void()
cartography$percentage_mean5 <- apply(ordinal_post, 2:3, mean)[, 5] * 100
p_percentage_mean5 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean5), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  theme_void()

# Sd
cartography$percentage_sd1 <- apply(ordinal_post, 2:3, sd)[, 1] * 100
p_percentage_sd1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
  theme_void()
cartography$percentage_sd2 <- apply(ordinal_post, 2:3, sd)[, 2] * 100
p_percentage_sd2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
  theme_void()
cartography$percentage_sd3 <- apply(ordinal_post, 2:3, sd)[, 3] * 100
p_percentage_sd3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
  theme_void()
cartography$percentage_sd4 <- apply(ordinal_post, 2:3, sd)[, 4] * 100
p_percentage_sd4 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd4), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
  theme_void()
cartography$percentage_sd5 <- apply(ordinal_post, 2:3, sd)[, 5] * 100
p_percentage_sd5 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd5), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
```

```
  theme_void()

# CV
cartography$percentage_CV1 <- 100 * cartography$percentage_sd1/cartography$percentage_mean1
p_percentage_CV1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  theme_void()
cartography$percentage_CV2 <- 100 * cartography$percentage_sd2/cartography$percentage_mean2
p_percentage_CV2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  theme_void()
cartography$percentage_CV3 <- 100 * cartography$percentage_sd3/cartography$percentage_mean3
p_percentage_CV3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  theme_void()
cartography$percentage_CV4 <- 100 * cartography$percentage_sd4/cartography$percentage_mean4
p_percentage_CV4 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV4), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  theme_void()
cartography$percentage_CV5 <- 100 * cartography$percentage_sd5/cartography$percentage_mean5
p_percentage_CV5 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV5), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  theme_void()

# Probability above Western European mean
we_mean1 <- apply(ordinal_post[, , 1], 1, mean)
we_mean2 <- apply(ordinal_post[, , 2], 1, mean)
we_mean3 <- apply(ordinal_post[, , 3], 1, mean)
we_mean4 <- apply(ordinal_post[, , 4], 1, mean)
we_mean5 <- apply(ordinal_post[, , 5], 1, mean)

we_mean_post1 <- mean(we_mean1) * 100
we_mean_post2 <- mean(we_mean2) * 100
we_mean_post3 <- mean(we_mean3) * 100
we_mean_post4 <- mean(we_mean4) * 100
we_mean_post5 <- mean(we_mean5) * 100

cartography$percentage_prob_above1 <- apply(ordinal_post[, , 1] > we_mean1, 2, mean)
cartography$percentage_prob_above2 <- apply(ordinal_post[, , 2] > we_mean2, 2, mean)
cartography$percentage_prob_above3 <- apply(ordinal_post[, , 3] > we_mean3, 2, mean)
cartography$percentage_prob_above4 <- apply(ordinal_post[, , 4] > we_mean4, 2, mean)
cartography$percentage_prob_above5 <- apply(ordinal_post[, , 5] > we_mean5, 2, mean)

cartography$percentage_low1 <- apply(ordinal_post[, , 1] * 100, 2, quantile, probs = 0.025)
cartography$percentage_low2 <- apply(ordinal_post[, , 2] * 100, 2, quantile, probs = 0.025)
cartography$percentage_low3 <- apply(ordinal_post[, , 3] * 100, 2, quantile, probs = 0.025)
cartography$percentage_low4 <- apply(ordinal_post[, , 4] * 100, 2, quantile, probs = 0.025)
cartography$percentage_low5 <- apply(ordinal_post[, , 5] * 100, 2, quantile, probs = 0.025)

cartography$percentage_up1 <- apply(ordinal_post[, , 1] * 100, 2, quantile, probs = 0.975)
cartography$percentage_up2 <- apply(ordinal_post[, , 2] * 100, 2, quantile, probs = 0.975)
cartography$percentage_up3 <- apply(ordinal_post[, , 3] * 100, 2, quantile, probs = 0.975)
cartography$percentage_up4 <- apply(ordinal_post[, , 4] * 100, 2, quantile, probs = 0.975)
cartography$percentage_up5 <- apply(ordinal_post[, , 5] * 100, 2, quantile, probs = 0.975)
```

```r
cartography$border_prob_above1 <- NA
cartography$border_prob_above1[cartography$percentage_low1 > we_mean_post1] <- "#67000D"
cartography$border_prob_above1[cartography$percentage_up1 < we_mean_post1] <- "#08306B"

cartography$border_prob_above2 <- NA
cartography$border_prob_above2[cartography$percentage_low2 > we_mean_post2] <- "#67000D"
cartography$border_prob_above2[cartography$percentage_up2 < we_mean_post2] <- "#08306B"

cartography$border_prob_above3 <- NA
cartography$border_prob_above3[cartography$percentage_low3 > we_mean_post3] <- "#67000D"
cartography$border_prob_above3[cartography$percentage_up3 < we_mean_post3] <- "#08306B"

cartography$border_prob_above4 <- NA
cartography$border_prob_above4[cartography$percentage_low4 > we_mean_post4] <- "#67000D"
cartography$border_prob_above4[cartography$percentage_up4 < we_mean_post4] <- "#08306B"

cartography$border_prob_above5 <- NA
cartography$border_prob_above5[cartography$percentage_low5 > we_mean_post5] <- "#67000D"
cartography$border_prob_above5[cartography$percentage_up5 < we_mean_post5] <- "#08306B"

p_percentage_prob_above1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above1, color = border_prob_above1), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_percentage_prob_above2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above2, color = border_prob_above2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_percentage_prob_above3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above3, color = border_prob_above3), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_percentage_prob_above4 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above4, color = border_prob_above4), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_percentage_prob_above5 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above5, color = border_prob_above5), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_percentage_mean1 <- p_percentage_mean1 + ggtitle(y_labels[1])
p_percentage_mean2 <- p_percentage_mean2 + ggtitle(y_labels[2])
p_percentage_mean3 <- p_percentage_mean3 + ggtitle(y_labels[3])
p_percentage_mean4 <- p_percentage_mean4 + ggtitle(y_labels[4])
p_percentage_mean5 <- p_percentage_mean5 + ggtitle(y_labels[5])

p_percentage_mean1 <- p_percentage_mean1 + labs(tag = "Mean")
p_percentage_sd1   <- p_percentage_sd1 + labs(tag = "SD")
```

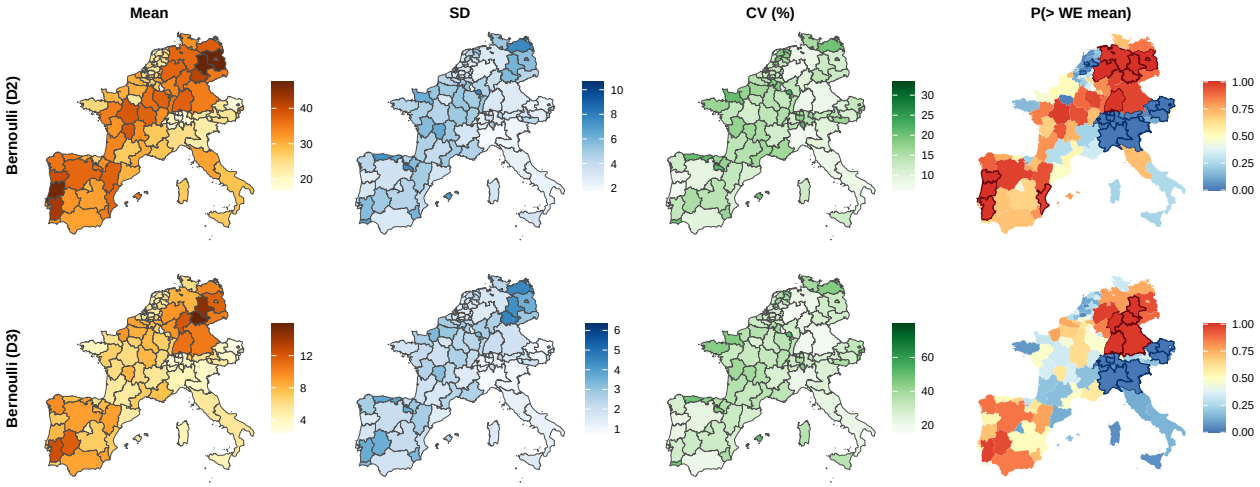
```
p_percentage_CV1   <- p_percentage_CV1 + labs(tag = "CV (%)")
p_percentage_prob_above1 <- p_percentage_prob_above1 + labs(tag = "P(> WE mean)")

tema_mapas <- theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
                    plot.tag = element_text(face = "bold", size = 12, angle = 90),
                    plot.tag.position = c(-0.08, 0.5),
                    plot.margin = margin(5.5, 5.5, 5.5, 20))

final_plot <- wrap_plots(p_percentage_mean1 + tema_mapas,
                         p_percentage_mean2 + tema_mapas,
                         p_percentage_mean3 + tema_mapas,
                         p_percentage_mean4 + tema_mapas,
                         p_percentage_mean5 + tema_mapas,
                         p_percentage_sd1 + tema_mapas,
                         p_percentage_sd2 + tema_mapas,
                         p_percentage_sd3 + tema_mapas,
                         p_percentage_sd4 + tema_mapas,
                         p_percentage_sd5 + tema_mapas,
                         p_percentage_CV1 + tema_mapas,
                         p_percentage_CV2 + tema_mapas,
                         p_percentage_CV3 + tema_mapas,
                         p_percentage_CV4 + tema_mapas,
                         p_percentage_CV5 + tema_mapas,
                         p_percentage_prob_above1 + tema_mapas,
                         p_percentage_prob_above2 + tema_mapas,
                         p_percentage_prob_above3 + tema_mapas,
                         p_percentage_prob_above4 + tema_mapas,
                         p_percentage_prob_above5 + tema_mapas, ncol = 5)
final_plot
```




```
### Bernoulli models: D+ and D- ###
```

```r
tema_mapas_bern <- theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
                         plot.tag = element_text(face = "bold", size = 12, angle = 90),
                         plot.tag.position = c(-0.08, 0.5),
                         plot.margin = margin(5.5, 5.5, 5.5, 20))

### Bernoulli (D2) ###

cartography$percentage_mean_D2 <- apply(bernoulli1_post, 2, mean) * 100
p_percentage_mean_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean_D2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  ggtitle("Mean") + theme_void()

cartography$percentage_sd_D2 <- apply(bernoulli1_post, 2, sd) * 100
p_percentage_sd_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd_D2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
  ggtitle("SD") + theme_void()

cartography$percentage_CV_D2 <- 100 * cartography$percentage_sd_D2 /
↪  cartography$percentage_mean_D2
p_percentage_CV_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV_D2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  ggtitle("CV (%)") + theme_void()

we_mean_D2 <- apply(bernoulli1_post, 1, mean)
we_mean_post_D2 <- mean(we_mean_D2) * 100

cartography$percentage_prob_above_D2 <- apply(bernoulli1_post > we_mean_D2, 2, mean)
cartography$percentage_low_D2 <- apply(bernoulli1_post * 100, 2, quantile, probs = 0.025)
cartography$percentage_up_D2 <- apply(bernoulli1_post * 100, 2, quantile, probs = 0.975)

cartography$border_prob_above_D2 <- NA
cartography$border_prob_above_D2[cartography$percentage_low_D2 > we_mean_post_D2] <- "#67000D"
cartography$border_prob_above_D2[cartography$percentage_up_D2 < we_mean_post_D2] <- "#08306B"

p_percentage_prob_above_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above_D2, color = border_prob_above_D2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + ggtitle("P(> WE mean)") + theme_void()

p_percentage_mean_D2 <- p_percentage_mean_D2 + labs(tag = "Bernoulli (D2)")

### Bernoulli (D3) ###

cartography$percentage_mean_D3 <- apply(bernoulli2_post, 2, mean) * 100
p_percentage_mean_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean_D3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"), name = NULL) +
  theme_void()

cartography$percentage_sd_D3 <- apply(bernoulli2_post, 2, sd) * 100
p_percentage_sd_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd_D3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), name = NULL) +
  theme_void()
```

```r
cartography$percentage_CV_D3 <- 100 * cartography$percentage_sd_D3 /
↪  cartography$percentage_mean_D3
p_percentage_CV_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV_D3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"), name = NULL) +
  theme_void()

we_mean_D3 <- apply(bernoulli2_post, 1, mean)
we_mean_post_D3 <- mean(we_mean_D3) * 100

cartography$percentage_prob_above_D3 <- apply(bernoulli2_post > we_mean_D3, 2, mean)
cartography$percentage_low_D3 <- apply(bernoulli2_post * 100, 2, quantile, probs = 0.025)
cartography$percentage_up_D3 <- apply(bernoulli2_post * 100, 2, quantile, probs = 0.975)

cartography$border_prob_above_D3 <- NA
cartography$border_prob_above_D3[cartography$percentage_low_D3 > we_mean_post_D3] <- "#67000D"
cartography$border_prob_above_D3[cartography$percentage_up_D3 < we_mean_post_D3] <- "#08306B"

p_percentage_prob_above_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_prob_above_D3, color = border_prob_above_D3), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_percentage_mean_D3 <- p_percentage_mean_D3 + labs(tag = "Bernoulli (D3)")

final_plot_bernoulli <- wrap_plots(p_percentage_mean_D2 + tema_mapas_bern,
                                   p_percentage_sd_D2 + tema_mapas_bern,
                                   p_percentage_CV_D2 + tema_mapas_bern,
                                   p_percentage_prob_above_D2 + tema_mapas_bern,
                                   p_percentage_mean_D3 + tema_mapas_bern,
                                   p_percentage_sd_D3 + tema_mapas_bern,
                                   p_percentage_CV_D3 + tema_mapas_bern,
                                   p_percentage_prob_above_D3 + tema_mapas_bern,
                                   ncol = 4)

final_plot_bernoulli
```



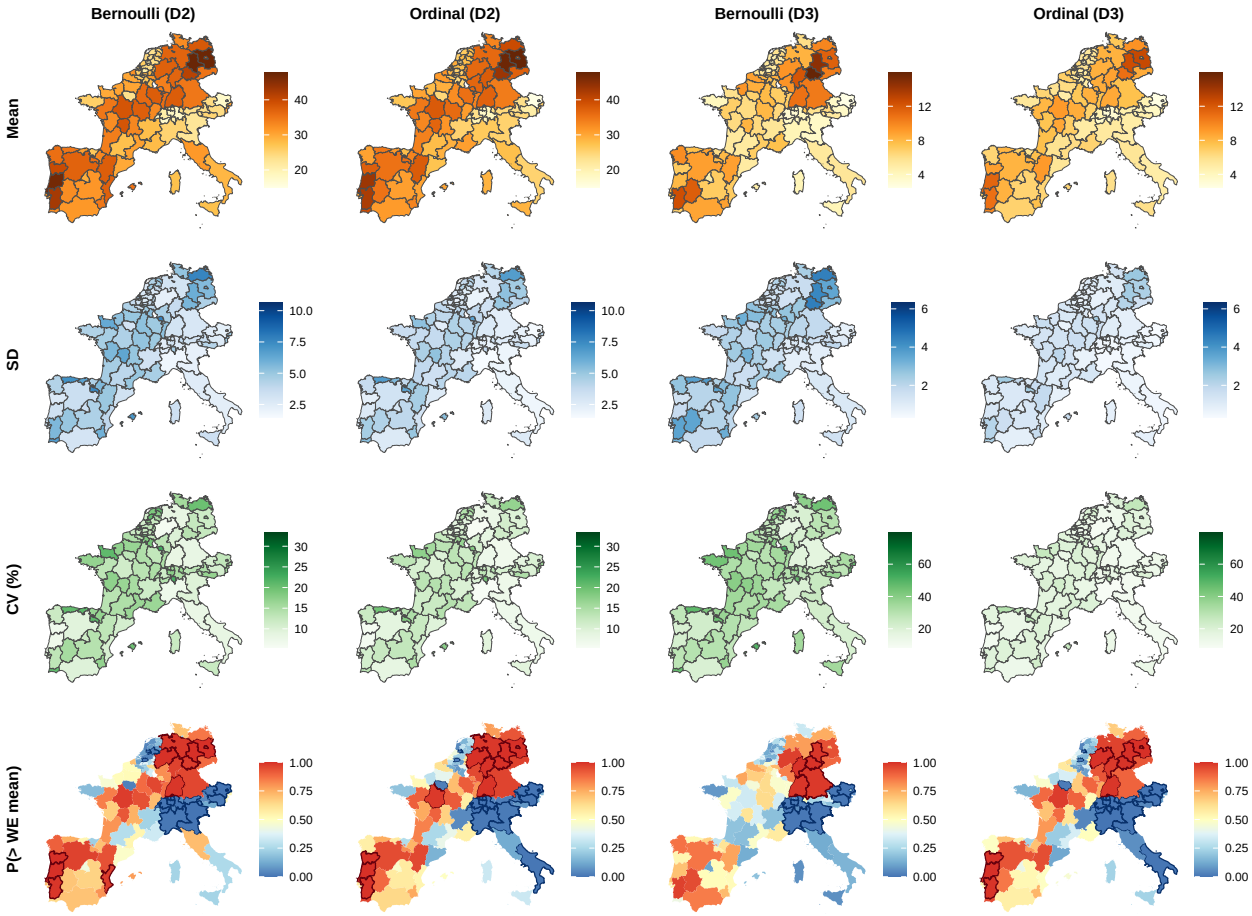
Comparison: Bernoulli and aggregated ordinal models

```r
# Ordinal aggregations
ordinal_D2_post <- ordinal_post[, , 3] + ordinal_post[, , 4] + ordinal_post[, , 5]
ordinal_D3_post <- ordinal_post[, , 4] + ordinal_post[, , 5]

# Mean
cartography$percentage_mean_bernoulli1 <- apply(bernoulli1_post, 2, mean) * 100
cartography$percentage_mean_ordinal_D2 <- apply(ordinal_D2_post, 2, mean) * 100
cartography$percentage_mean_bernoulli2 <- apply(bernoulli2_post, 2, mean) * 100
cartography$percentage_mean_ordinal_D3 <- apply(ordinal_D3_post, 2, mean) * 100

# Sd
cartography$percentage_sd_bernoulli1 <- apply(bernoulli1_post, 2, sd) * 100
cartography$percentage_sd_ordinal_D2 <- apply(ordinal_D2_post, 2, sd) * 100
cartography$percentage_sd_bernoulli2 <- apply(bernoulli2_post, 2, sd) * 100
cartography$percentage_sd_ordinal_D3 <- apply(ordinal_D3_post, 2, sd) * 100

# CV
cartography$percentage_CV_bernoulli1 <- 100 * cartography$percentage_sd_bernoulli1 /
↪  cartography$percentage_mean_bernoulli1
cartography$percentage_CV_ordinal_D2 <- 100 * cartography$percentage_sd_ordinal_D2 /
↪  cartography$percentage_mean_ordinal_D2
cartography$percentage_CV_bernoulli2 <- 100 * cartography$percentage_sd_bernoulli2 /
↪  cartography$percentage_mean_bernoulli2
cartography$percentage_CV_ordinal_D3 <- 100 * cartography$percentage_sd_ordinal_D3 /
↪  cartography$percentage_mean_ordinal_D3

# Common scales within each comparison
limit_mean_D2 <- range(c(cartography$percentage_mean_bernoulli1,
                         cartography$percentage_mean_ordinal_D2), na.rm = TRUE)
limit_mean_D3 <- range(c(cartography$percentage_mean_bernoulli2,
                          cartography$percentage_mean_ordinal_D3), na.rm = TRUE)

limit_sd_D2 <- range(c(cartography$percentage_sd_bernoulli1,
                       cartography$percentage_sd_ordinal_D2), na.rm = TRUE)
limit_sd_D3 <- range(c(cartography$percentage_sd_bernoulli2,
                        cartography$percentage_sd_ordinal_D3), na.rm = TRUE)

limit_CV_D2 <- range(c(cartography$percentage_CV_bernoulli1,
                       cartography$percentage_CV_ordinal_D2), na.rm = TRUE)
limit_CV_D3 <- range(c(cartography$percentage_CV_bernoulli2,
                        cartography$percentage_CV_ordinal_D3), na.rm = TRUE)

# Mean maps
p_percentage_mean_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean_bernoulli1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"),
                       limits = limit_mean_D2,
                       name = NULL) + ggtitle("Bernoulli (D2)") + theme_void()

p_percentage_mean_ordinal_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean_ordinal_D2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"),
                       limits = limit_mean_D2,
                       name = NULL) + ggtitle("Ordinal (D2)") + theme_void()

p_percentage_mean_bernoulli2 <- ggplot(cartography) +
```

```
  geom_sf(aes(fill = percentage_mean_bernoulli2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"),
                       limits = limit_mean_D3,
                       name = NULL) + ggtitle("Bernoulli (D3)") + theme_void()

p_percentage_mean_ordinal_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_mean_ordinal_D3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "YlOrBr"),
                       limits = limit_mean_D3,
                       name = NULL) + ggtitle("Ordinal (D3)") + theme_void()

# Sd maps
p_percentage_sd_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd_bernoulli1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = limit_sd_D2,
                       name = NULL) + theme_void()

p_percentage_sd_ordinal_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd_ordinal_D2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = limit_sd_D2,
                       name = NULL) + theme_void()

p_percentage_sd_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd_bernoulli2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = limit_sd_D3,
                       name = NULL) + theme_void()

p_percentage_sd_ordinal_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_sd_ordinal_D3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"),
                       limits = limit_sd_D3,
                       name = NULL) + theme_void()

# CV maps
p_percentage_CV_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV_bernoulli1), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"),
                       limits = limit_CV_D2,
                       name = NULL) +  theme_void()

p_percentage_CV_ordinal_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV_ordinal_D2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"),
                       limits = limit_CV_D2,
                       name = NULL) + theme_void()

p_percentage_CV_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV_bernoulli2), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"),
                       limits = limit_CV_D3,
                       name = NULL) + theme_void()

p_percentage_CV_ordinal_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = percentage_CV_ordinal_D3), color = "grey30", linewidth = 0.1) +
  scale_fill_gradientn(colours = brewer.pal(9, "Greens"),
                       limits = limit_CV_D3,
```

```
                      name = NULL) + theme_void()

# Probability above Western European mean

we_mean_bernoulli1 <- apply(bernoulli1_post, 1, mean)
we_mean_ordinal_D2 <- apply(ordinal_D2_post, 1, mean)
we_mean_bernoulli2 <- apply(bernoulli2_post, 1, mean)
we_mean_ordinal_D3 <- apply(ordinal_D3_post, 1, mean)

we_mean_post_bernoulli1 <- mean(we_mean_bernoulli1) * 100
we_mean_post_ordinal_D2 <- mean(we_mean_ordinal_D2) * 100
we_mean_post_bernoulli2 <- mean(we_mean_bernoulli2) * 100
we_mean_post_ordinal_D3 <- mean(we_mean_ordinal_D3) * 100

cartography$prob_above_bernoulli1 <- apply(bernoulli1_post > we_mean_bernoulli1, 2, mean)
cartography$prob_above_ordinal_D2 <- apply(ordinal_D2_post > we_mean_ordinal_D2, 2, mean)
cartography$prob_above_bernoulli2 <- apply(bernoulli2_post > we_mean_bernoulli2, 2, mean)
cartography$prob_above_ordinal_D3 <- apply(ordinal_D3_post > we_mean_ordinal_D3, 2, mean)

cartography$low_bernoulli1 <- apply(bernoulli1_post * 100, 2, quantile, probs = 0.025)
cartography$up_bernoulli1 <- apply(bernoulli1_post * 100, 2, quantile, probs = 0.975)

cartography$low_ordinal_D2 <- apply(ordinal_D2_post * 100, 2, quantile, probs = 0.025)
cartography$up_ordinal_D2 <- apply(ordinal_D2_post * 100, 2, quantile, probs = 0.975)

cartography$low_bernoulli2 <- apply(bernoulli2_post * 100, 2, quantile, probs = 0.025)
cartography$up_bernoulli2 <- apply(bernoulli2_post * 100, 2, quantile, probs = 0.975)

cartography$low_ordinal_D3 <- apply(ordinal_D3_post * 100, 2, quantile, probs = 0.025)
cartography$up_ordinal_D3 <- apply(ordinal_D3_post * 100, 2, quantile, probs = 0.975)

cartography$border_above_bernoulli1 <- NA
cartography$border_above_bernoulli1[cartography$low_bernoulli1 > we_mean_post_bernoulli1] <-
↪  "#67000D"
cartography$border_above_bernoulli1[cartography$up_bernoulli1 < we_mean_post_bernoulli1] <-
↪  "#08306B"

cartography$border_above_ordinal_D2 <- NA
cartography$border_above_ordinal_D2[cartography$low_ordinal_D2 > we_mean_post_ordinal_D2] <-
↪  "#67000D"
cartography$border_above_ordinal_D2[cartography$up_ordinal_D2 < we_mean_post_ordinal_D2] <-
↪  "#08306B"

cartography$border_above_bernoulli2 <- NA
cartography$border_above_bernoulli2[cartography$low_bernoulli2 > we_mean_post_bernoulli2] <-
↪  "#67000D"
cartography$border_above_bernoulli2[cartography$up_bernoulli2 < we_mean_post_bernoulli2] <-
↪  "#08306B"

cartography$border_above_ordinal_D3 <- NA
cartography$border_above_ordinal_D3[cartography$low_ordinal_D3 > we_mean_post_ordinal_D3] <-
↪  "#67000D"
cartography$border_above_ordinal_D3[cartography$up_ordinal_D3 < we_mean_post_ordinal_D3] <-
↪  "#08306B"

p_prob_above_bernoulli1 <- ggplot(cartography) +
  geom_sf(aes(fill = prob_above_bernoulli1, color = border_above_bernoulli1), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
```

```r
  scale_color_identity(na.value = NA) + theme_void()

p_prob_above_ordinal_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = prob_above_ordinal_D2, color = border_above_ordinal_D2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_prob_above_bernoulli2 <- ggplot(cartography) +
  geom_sf(aes(fill = prob_above_bernoulli2, color = border_above_bernoulli2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

p_prob_above_ordinal_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = prob_above_ordinal_D3, color = border_above_ordinal_D3), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

# Row labels
p_percentage_mean_bernoulli1 <- p_percentage_mean_bernoulli1 + labs(tag = "Mean")
p_percentage_sd_bernoulli1 <- p_percentage_sd_bernoulli1 + labs(tag = "SD")
p_percentage_CV_bernoulli1 <- p_percentage_CV_bernoulli1 + labs(tag = "CV (%)")
p_prob_above_bernoulli1 <- p_prob_above_bernoulli1 + labs(tag = "P(> WE mean)")

# Theme
tema_mapas <- theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
                    plot.tag = element_text(face = "bold", size = 12, angle = 90),
                    plot.tag.position = c(-0.08, 0.5), plot.margin = margin(5.5, 5.5, 5.5, 20))

final_plot <- wrap_plots(p_percentage_mean_bernoulli1 + tema_mapas,
                         p_percentage_mean_ordinal_D2 + tema_mapas,
                         p_percentage_mean_bernoulli2 + tema_mapas,
                         p_percentage_mean_ordinal_D3 + tema_mapas,
                         p_percentage_sd_bernoulli1 + tema_mapas,
                         p_percentage_sd_ordinal_D2 + tema_mapas,
                         p_percentage_sd_bernoulli2 + tema_mapas,
                         p_percentage_sd_ordinal_D3 + tema_mapas,
                         p_percentage_CV_bernoulli1 + tema_mapas,
                         p_percentage_CV_ordinal_D2 + tema_mapas,
                         p_percentage_CV_bernoulli2 + tema_mapas,
                         p_percentage_CV_ordinal_D3 + tema_mapas,
                         p_prob_above_bernoulli1 + tema_mapas,
                         p_prob_above_ordinal_D2 + tema_mapas,
                         p_prob_above_bernoulli2 + tema_mapas,
                         p_prob_above_ordinal_D3 + tema_mapas, ncol = 4)

final_plot
```

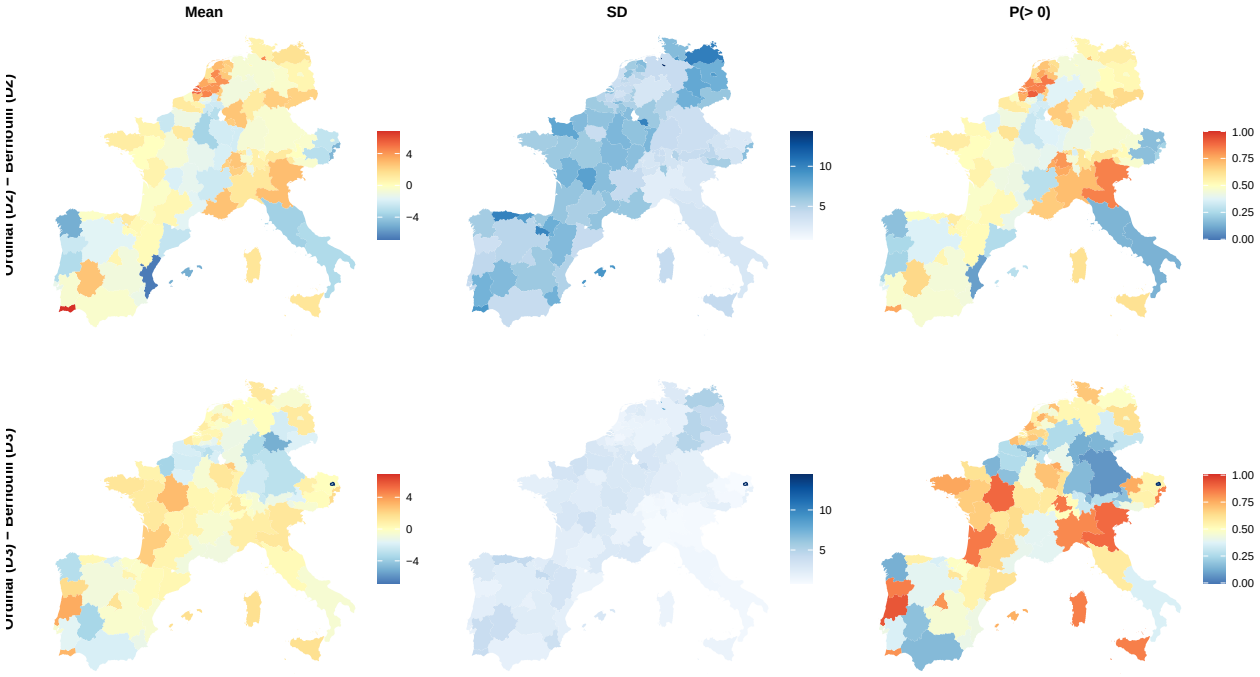


Differences: Bernoulli and aggregated ordinal models

```
ordinal_D2_post <- ordinal_post[, , 3] + ordinal_post[, , 4] + ordinal_post[, , 5]
ordinal_D3_post <- ordinal_post[, , 4] + ordinal_post[, , 5]

diff_D2_post <- ordinal_D2_post - bernoulli1_post
diff_D3_post <- ordinal_D3_post - bernoulli2_post

# Mean
cartography$diffmean_D2 <- apply(diff_D2_post, 2, mean) * 100
cartography$diffmean_D3 <- apply(diff_D3_post, 2, mean) * 100

# Sd
cartography$diffsd_D2 <- apply(diff_D2_post, 2, sd) * 100
cartography$diffsd_D3 <- apply(diff_D3_post, 2, sd) * 100

# 95% credible intervals for the differences
cartography$difflow_D2 <- apply(diff_D2_post * 100, 2, quantile, probs = 0.025)
cartography$diffup_D2  <- apply(diff_D2_post * 100, 2, quantile, probs = 0.975)

cartography$difflow_D3 <- apply(diff_D3_post * 100, 2, quantile, probs = 0.025)
cartography$diffup_D3  <- apply(diff_D3_post * 100, 2, quantile, probs = 0.975)
```

```r
# Probability that the difference is greater than zero
D2_stepsim <- 1 * (diff_D2_post > 0)
D3_stepsim <- 1 * (diff_D3_post > 0)

cartography$prob_diff_D2_great0 <- apply(D2_stepsim, 2, mean)
cartography$prob_diff_D3_great0 <- apply(D3_stepsim, 2, mean)

cartography$problow_D2 <- apply(D2_stepsim, 2, quantile, probs = 0.025)
cartography$probup_D2  <- apply(D2_stepsim, 2, quantile, probs = 0.975)

cartography$problow_D3 <- apply(D3_stepsim, 2, quantile, probs = 0.025)
cartography$probup_D3  <- apply(D3_stepsim, 2, quantile, probs = 0.975)

# Borders
cartography$border_mean_D2 <- NA
cartography$border_mean_D2[cartography$difflow_D2 > 0] <- "#67000D"
cartography$border_mean_D2[cartography$diffup_D2 < 0] <- "#08306B"

cartography$border_mean_D3 <- NA
cartography$border_mean_D3[cartography$difflow_D3 > 0] <- "#67000D"
cartography$border_mean_D3[cartography$diffup_D3 < 0] <- "#08306B"

cartography$border_sd_D2 <- NA
cartography$border_sd_D2[cartography$difflow_D2 > 0 | cartography$diffup_D2 < 0] <- "#08306B"

cartography$border_sd_D3 <- NA
cartography$border_sd_D3[cartography$difflow_D3 > 0 | cartography$diffup_D3 < 0] <- "#08306B"

cartography$border_prob_D2 <- NA
cartography$border_prob_D2[cartography$problow_D2 > 0.5] <- "#67000D"
cartography$border_prob_D2[cartography$probup_D2 < 0.5] <- "#08306B"

cartography$border_prob_D3 <- NA
cartography$border_prob_D3[cartography$problow_D3 > 0.5] <- "#67000D"
cartography$border_prob_D3[cartography$probup_D3 < 0.5] <- "#08306B"

# Common scales
limit_diffmean <- max(abs(c(cartography$diffmean_D2,
                            cartography$diffmean_D3)), na.rm = TRUE)

limit_diffsd <- range(c(cartography$diffsd_D2,
                        cartography$diffsd_D3), na.rm = TRUE)

# Mean maps
p_diffmean_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = diffmean_D2, color = border_mean_D2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(-limit_diffmean, limit_diffmean),
                       values = rescale(c(-limit_diffmean, 0, limit_diffmean)),
                       name = NULL) +
  scale_color_identity(na.value = NA) + ggtitle("Mean") + theme_void()

p_diffmean_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = diffmean_D3, color = border_mean_D3), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(-limit_diffmean, limit_diffmean),
                       values = rescale(c(-limit_diffmean, 0, limit_diffmean)),
                       name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()
```

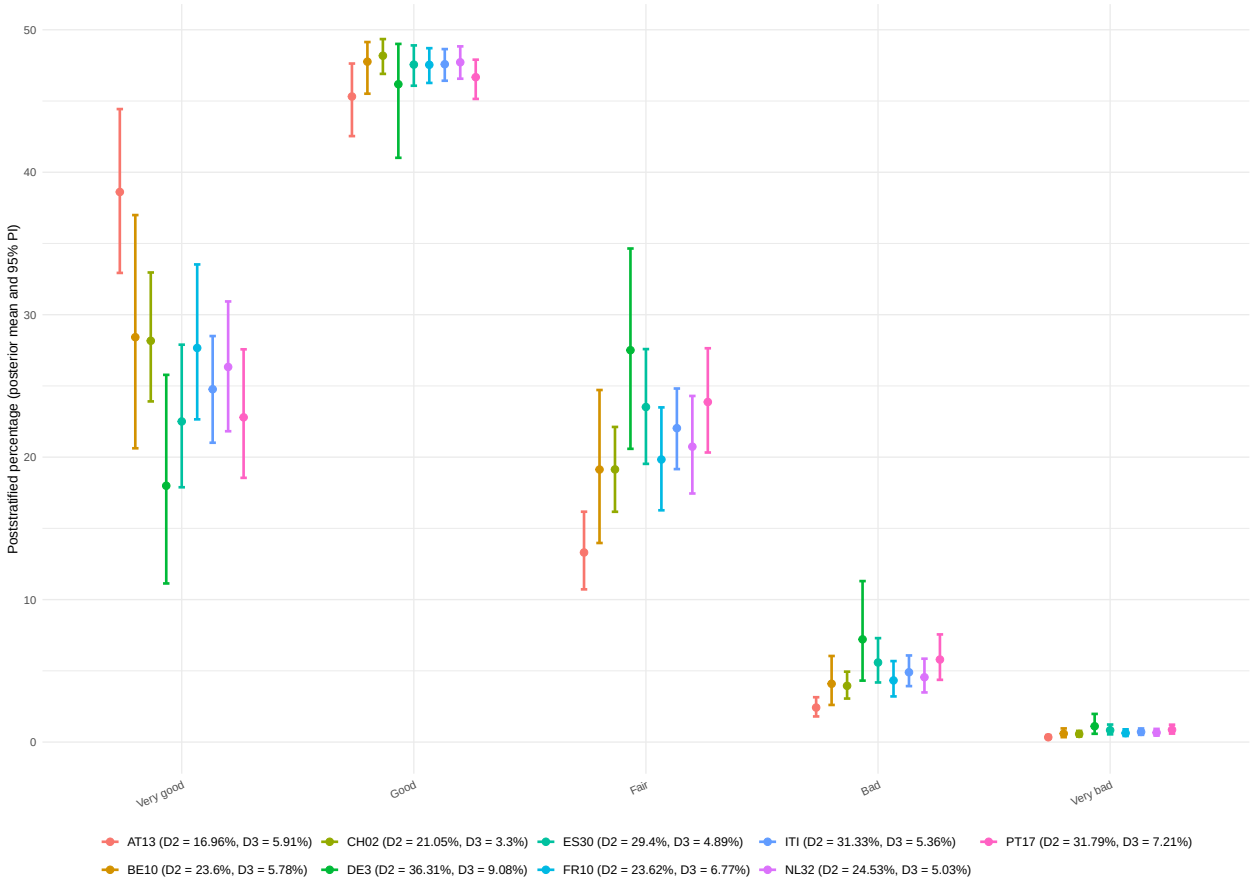
```r
# Sd maps
p_diffsd_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = diffsd_D2, color = border_sd_D2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), limits = limit_diffsd, name = NULL) +
  scale_color_identity(na.value = NA) + ggtitle("SD") + theme_void()

p_diffsd_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = diffsd_D3, color = border_sd_D3), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "Blues"), limits = limit_diffsd, name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

# Probability maps
p_prob_D2 <- ggplot(cartography) +
  geom_sf(aes(fill = prob_diff_D2_great0, color = border_prob_D2), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + ggtitle("P(> 0)") + theme_void()

p_prob_D3 <- ggplot(cartography) +
  geom_sf(aes(fill = prob_diff_D3_great0, color = border_prob_D3), linewidth = 0.45) +
  scale_fill_gradientn(colours = brewer.pal(9, "RdYlBu")[9:1],
                       limits = c(0, 1), values = rescale(c(0, 0.5, 1)), name = NULL) +
  scale_color_identity(na.value = NA) + theme_void()

# Row labels
p_diffmean_D2 <- p_diffmean_D2 + labs(tag = "Ordinal (D2) - Bernoulli (D2)")
p_diffmean_D3 <- p_diffmean_D3 + labs(tag = "Ordinal (D3) - Bernoulli (D3)")

# Theme
tema_mapas_diff <- theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
                         plot.tag = element_text(face = "bold", size = 12, angle = 90),
                         plot.tag.position = c(-0.08, 0.5),
                         plot.margin = margin(5.5, 5.5, 5.5, 20))

# Final plot
final_plot_diff <- wrap_plots(p_diffmean_D2 + tema_mapas_diff,
                              p_diffsd_D2 + tema_mapas_diff,
                              p_prob_D2 + tema_mapas_diff,
                              p_diffmean_D3 + tema_mapas_diff,
                              p_diffsd_D3 + tema_mapas_diff,
                              p_prob_D3 + tema_mapas_diff, ncol = 3)

final_plot_diff
```



Ordinal profiles in selected regions

```
capital_nuts <- c("AT13", "BE10", "CH02", "DE3", "ES30",
                  "FR10", "ITI", "NL32", "PT17")

selected_ids <- match(capital_nuts, cartography$NUTS_ID)

df_capitals <- data.frame()
for (capital in selected_ids) {

  ordinal_area <- ordinal_post[, capital, ] * 100
  bernoulli1_area <- apply(bernoulli1_post, 2, mean)[capital] * 100
  bernoulli2_area <- apply(bernoulli2_post, 2, mean)[capital] * 100

  df_capitals <- rbind(df_capitals,data.frame(
    category = c("Very good", "Good", "Fair", "Bad", "Very bad"),
    mean = apply(ordinal_area, 2, mean),
    lower = apply(ordinal_area, 2, quantile, probs = 0.025),
    upper = apply(ordinal_area, 2, quantile, probs = 0.975),
    area = paste0(cartography$NUTS_ID[capital],
                  " (D2 = ", round(bernoulli1_area, 2), "%",
                  ", D3 = ", round(bernoulli2_area, 2), "%)")))
}

df_capitals$category <- factor(df_capitals$category,
                               levels = c("Very good", "Good", "Fair", "Bad", "Very bad"))

p_capitals <- ggplot(df_capitals, aes(x = category, y = mean, color = area)) +
  geom_point(position = position_dodge(width = 0.6), size = 2.5) +
  geom_errorbar(aes(ymin = lower, ymax = upper),
                position = position_dodge(width = 0.6), width = 0.25, linewidth = 1) +
  theme_minimal() + labs(x = NULL, y = "Poststratified percentage (posterior mean and 95% PI)",
  ↪  color = NULL) +
```

```
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
        axis.text.x = element_text(angle = 25, hjust = 1),
        legend.position = "bottom", legend.text = element_text(size = 10.25),
        legend.key.size = unit(0.6, "cm"))

p_capitals
```



Convergence assessment

```
### Ordinal model ###

round(ordinal_results$summary, 4)
```

```
##                mean     sd    2.5%     50%   97.5% Rhat n.eff
## beta_age[1]  0.0000 0.0000  0.0000  0.0000  0.0000  NaN     0
## beta_age[2]  0.7874 0.0607  0.6586  0.7843  0.9083 1.01  1274
## beta_age[3]  1.7564 0.0580  1.6378  1.7588  1.8673 1.00  1125
## beta_edu[1]  0.0000 0.0000  0.0000  0.0000  0.0000  NaN     0
## beta_edu[2] -0.4354 0.0576 -0.5512 -0.4347 -0.3224 1.01   951
## beta_edu[3] -0.8457 0.0596 -0.9654 -0.8452 -0.7301 1.01  1000
## delta[1]     0.3179 0.0130  0.2913  0.3185  0.3431 1.00  1044
```

```
## delta[2]     0.5155 0.0072  0.5016  0.5155  0.5293 1.01  1000
## delta[3]     0.1399 0.0074  0.1266  0.1396  0.1548 1.00  1058
## delta[4]     0.0234 0.0018  0.0203  0.0233  0.0272 1.00  1000
## delta[5]     0.0032 0.0004  0.0024  0.0032  0.0042 1.00  1045
## kappa[1]    -0.7640 0.0600 -0.8892 -0.7609 -0.6493 1.00  1043
## kappa[2]     1.6116 0.0640  1.4837  1.6119  1.7287 1.00  1026
## kappa[3]     3.6010 0.0752  3.4452  3.6020  3.7405 1.00  1000
## kappa[4]     5.7455 0.1362  5.4713  5.7395  6.0179 1.00  1047
## rho          0.6339 0.1528  0.3316  0.6482  0.9019 1.00  1037
## sd.theta     0.5232 0.0619  0.4133  0.5191  0.6527 1.00   835
## theta[1]    -0.2871 0.4646 -1.1842 -0.2893  0.6331 1.00  1118
## theta[2]    -1.5527 0.2836 -2.1323 -1.5409 -1.0213 1.00  1000
## theta[3]    -1.1531 0.3074 -1.7505 -1.1476 -0.6109 1.01   894
## theta[4]    -0.3437 0.4124 -1.1815 -0.3508  0.4664 1.00  1102
## theta[5]    -0.6914 0.2768 -1.2643 -0.6896 -0.1729 1.00   972
## theta[6]    -0.3487 0.2522 -0.8498 -0.3405  0.1087 1.01  1006
## theta[7]    -0.3091 0.3412 -0.9618 -0.3215  0.3759 1.00   939
## theta[8]    -0.2206 0.3119 -0.8320 -0.2188  0.3956 1.00   916
## theta[9]    -0.3228 0.3875 -1.0584 -0.3308  0.4193 1.01  1076
## theta[10]   -0.1339 0.4390 -1.0816 -0.1407  0.7298 1.01   959
## theta[11]   -0.3025 0.2369 -0.7736 -0.3055  0.1631 1.00  1000
## theta[12]   -0.3802 0.3319 -1.0121 -0.3853  0.2607 1.00  1208
## theta[13]    0.0452 0.2931 -0.5340  0.0559  0.5903 1.00  1062
## theta[14]    0.0535 0.2944 -0.5133  0.0678  0.6088 1.00  1077
## theta[15]    0.2376 0.3245 -0.3603  0.2405  0.8928 1.00   899
## theta[16]   -0.2703 0.4796 -1.1940 -0.2558  0.5914 1.01   992
## theta[17]   -0.2173 0.3287 -0.9014 -0.1995  0.4147 1.00  1083
## theta[18]    0.1105 0.3497 -0.5840  0.0981  0.7580 1.00  1209
## theta[19]   -0.3010 0.4849 -1.2121 -0.2921  0.5716 1.00   943
## theta[20]   -0.4263 0.4615 -1.2998 -0.4257  0.4231 1.01  1000
## theta[21]   -0.6578 0.2735 -1.2308 -0.6446 -0.1400 1.00  1000
## theta[22]   -0.2473 0.2414 -0.7220 -0.2408  0.2031 1.00  1941
## theta[23]   -0.5191 0.3264 -1.1705 -0.5305  0.1028 1.01  1055
## theta[24]   -0.6911 0.3279 -1.3560 -0.6713 -0.0967 1.00  1070
## theta[25]   -0.8635 0.2831 -1.4321 -0.8525 -0.3097 1.00  1107
## theta[26]   -0.9342 0.3371 -1.5962 -0.9232 -0.2952 1.01  1000
## theta[27]   -0.9399 0.5068 -1.9967 -0.9294 -0.0163 1.00   944
## theta[28]    1.0466 0.2634  0.5401  1.0416  1.5980 1.00  1000
## theta[29]    0.7915 0.2378  0.3526  0.7826  1.2667 1.00  1063
## theta[30]    1.0649 0.5690  0.0483  1.0616  2.2266 1.01   939
## theta[31]    1.7645 0.3929  1.0234  1.7473  2.5741 1.00  1101
## theta[32]    0.4997 1.0061 -1.4241  0.4927  2.5292 1.01  1000
## theta[33]    0.4234 0.5922 -0.7398  0.4309  1.5327 1.00  1418
## theta[34]    1.0647 0.3099  0.4648  1.0613  1.6999 1.01   956
## theta[35]    1.0288 0.5989 -0.1733  1.0481  2.1989 1.00   969
## theta[36]    0.8466 0.2723  0.3115  0.8419  1.3899 1.00  1110
## theta[37]    0.9051 0.2193  0.5115  0.8946  1.3596 1.01  1000
## theta[38]    0.8759 0.3656  0.1190  0.8671  1.6255 1.01  1000
## theta[39]    0.7550 0.6220 -0.4886  0.7657  1.9260 1.00  1066
## theta[40]    0.9629 0.3875  0.2740  0.9547  1.7496 1.00  1063
## theta[41]    1.6854 0.4487  0.7773  1.6848  2.5664 1.01  1000
## theta[42]    0.5705 0.4087 -0.1599  0.5510  1.4116 1.01  1146
## theta[43]    1.4231 0.4380  0.5870  1.4212  2.3122 1.00  1101
## theta[44]   -0.1054 0.3863 -0.8829 -0.0932  0.6413 1.00  1140
```

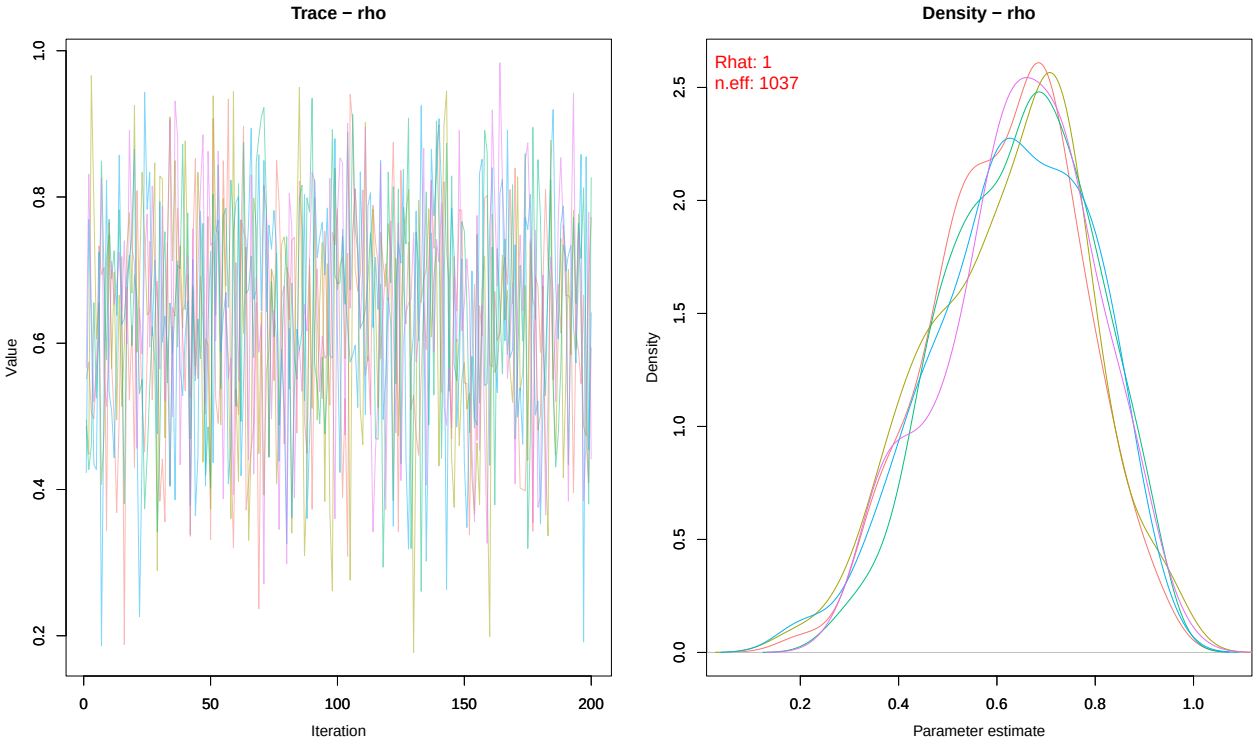
```
## theta[45]    0.3296 0.6604 -0.9415  0.3320  1.5571 1.00  1388
## theta[46]    0.1306 0.5853 -1.0208  0.1652  1.2098 1.00  1000
## theta[47]   -0.0613 0.4135 -0.8496 -0.0665  0.7510 1.00  1057
## theta[48]    0.4906 0.4925 -0.3807  0.4682  1.4391 1.01   971
## theta[49]    0.5284 0.6229 -0.7742  0.5130  1.7040 1.01  1000
## theta[50]    0.8031 0.4317 -0.0698  0.8106  1.6603 1.00  1049
## theta[51]    0.3787 0.3105 -0.2187  0.3791  0.9914 1.01  1046
## theta[52]    0.3561 0.3127 -0.2739  0.3593  0.9390 1.01  1090
## theta[53]    0.0347 0.4123 -0.7928  0.0522  0.8049 1.00  1045
## theta[54]    0.3754 0.4235 -0.4847  0.3813  1.2142 1.00  1004
## theta[55]   -0.0897 0.2906 -0.6660 -0.0870  0.4873 1.00   990
## theta[56]    0.0677 0.3361 -0.6198  0.0724  0.7029 1.00   955
## theta[57]    0.2063 0.5574 -0.8862  0.1888  1.2776 1.00  1045
## theta[58]    0.1818 0.2744 -0.3453  0.1666  0.7431 1.00  1052
## theta[59]    0.4854 0.5026 -0.5159  0.4828  1.4910 1.00  1052
## theta[60]    0.0636 0.2922 -0.5335  0.0603  0.6570 1.01   994
## theta[61]    0.9280 0.3520  0.2487  0.9444  1.5870 1.01  1097
## theta[62]    0.6787 0.4207 -0.1507  0.6771  1.5273 1.00  1000
## theta[63]    0.3119 0.4332 -0.5286  0.3094  1.1423 1.00  1373
## theta[64]    0.1174 0.5479 -0.9751  0.1286  1.2618 1.01   947
## theta[65]    0.0073 0.4784 -0.8891  0.0084  0.9762 1.00  1039
## theta[66]    0.5041 0.3995 -0.2447  0.5037  1.3265 1.00  1049
## theta[67]    0.1331 0.4218 -0.6960  0.1411  0.9196 1.00  1062
## theta[68]    0.7107 0.4345 -0.1433  0.7060  1.6196 1.00   852
## theta[69]    0.4382 0.4443 -0.4473  0.4593  1.3291 1.00  1005
## theta[70]    0.7963 0.3979  0.0287  0.8014  1.5744 1.00  1108
## theta[71]    0.7592 0.3717  0.0770  0.7345  1.4993 1.00  1008
## theta[72]   -0.0805 0.3954 -0.8087 -0.1033  0.7499 1.00  1682
## theta[73]    0.6103 0.3805 -0.0979  0.6109  1.3449 1.00   950
## theta[74]    0.6849 0.4891 -0.2905  0.6972  1.6355 1.00  1355
## theta[75]    0.3946 0.4530 -0.5087  0.3774  1.2427 1.00  1145
## theta[76]    0.0645 0.4347 -0.7968  0.0742  0.8664 1.00  1046
## theta[77]    0.0697 0.3743 -0.6424  0.0584  0.8361 1.00   923
## theta[78]    0.2895 0.4274 -0.5624  0.2902  1.0890 1.01  1680
## theta[79]   -0.1092 0.3001 -0.6880 -0.1047  0.4825 1.00  1118
## theta[80]    0.2776 0.4167 -0.5165  0.2861  1.0908 1.00   953
## theta[81]   -0.4912 0.1682 -0.8460 -0.4784 -0.1771 1.00   949
## theta[82]   -0.6117 0.2207 -1.0440 -0.6028 -0.2037 1.00  1125
## theta[83]   -0.2940 0.2886 -0.8842 -0.2889  0.2576 1.00  1115
## theta[84]   -0.6080 0.2268 -1.0662 -0.5892 -0.1702 1.01  1061
## theta[85]   -0.3364 0.2225 -0.7922 -0.3319  0.0692 1.00  1255
## theta[86]    0.0972 0.4815 -0.8689  0.1093  0.9902 1.00   997
## theta[87]   -0.2708 0.4172 -1.1588 -0.2706  0.5485 1.00  1210
## theta[88]   -0.1735 0.4356 -1.0312 -0.1813  0.7108 1.01  1059
## theta[89]   -0.0126 0.3438 -0.6878 -0.0070  0.6459 1.00  1583
## theta[90]    0.0209 0.2727 -0.5114  0.0164  0.5849 1.00  1000
## theta[91]    0.1493 0.4488 -0.7052  0.1464  0.9958 1.00  1008
## theta[92]   -0.4184 0.3591 -1.1045 -0.4061  0.2688 1.00  1097
## theta[93]   -0.0206 0.2610 -0.5139 -0.0247  0.4986 1.00  1000
## theta[94]    0.3193 0.2674 -0.2352  0.3242  0.8299 1.01  1089
## theta[95]    0.0570 0.4527 -0.7873  0.0513  1.0237 1.00  1004
## theta[96]    0.0364 0.2862 -0.5134  0.0336  0.6151 1.00  1114
## theta[97]    0.1864 0.3629 -0.5111  0.1844  0.8841 1.00   991
## theta[98]    0.4876 0.2334  0.0416  0.4864  0.9516 1.00   846
```

```
## theta[99]    0.6666 0.5080 -0.3362  0.6707  1.6763 1.00  1089
## theta[100]   0.9155 0.2674  0.4189  0.8977  1.4847 1.00  1227
## theta[101]   0.1262 0.2783 -0.4100  0.1254  0.6958 1.00  1000
## theta[102]   0.7779 0.4381 -0.0641  0.7606  1.6501 1.00  1000
```

```
MCMCsummary(object = ordinal_results$samples, params = "rho",
            # exact = TRUE,
            # ISB = FALSE,
            round = 4)
```

```
##       mean     sd   2.5%    50%  97.5% Rhat n.eff
## rho 0.6339 0.1528 0.3316 0.6482 0.9019    1  1037
```

```
MCMCtrace(object = ordinal_results$samples,
          pdf = FALSE, # no export to PDF
          ind = TRUE, # separate density lines per chain
          Rhat = TRUE,
          n.eff = TRUE,
          params = "rho")
```




```
test <- ordinal_results$samples

which((MCMCsummary(object = test, params = "kappa", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪  = test, params = "kappa", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "beta_age", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "beta_age", round = 4)[, 7] < 400))
```

```
## [1] 1
```

```
which((MCMCsummary(object = test, params = "beta_edu", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "beta_edu", round = 4)[, 7] < 400))
```

```
## [1] 1
```

```
which((MCMCsummary(object = test, params = "theta", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪  = test, params = "theta", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "sd.theta", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "sd.theta", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "rho", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪  = test, params = "rho", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
### Bernoulli (D+) model ###

round(bernoulli_results1$summary, 4)
```

```
##                mean     sd    2.5%     50%   97.5% Rhat n.eff
## beta_0      -1.4713 0.0794 -1.6355 -1.4722 -1.3183 1.01  1052
## beta_age[1]  0.0000 0.0000  0.0000  0.0000  0.0000  NaN     0
## beta_age[2]  0.6750 0.0874  0.5142  0.6742  0.8481 1.01  1045
## beta_age[3]  1.6856 0.0763  1.5370  1.6804  1.8386 1.01  1054
## beta_edu[1]  0.0000 0.0000  0.0000  0.0000  0.0000  NaN     0
## beta_edu[2] -0.5212 0.0680 -0.6562 -0.5202 -0.3930 1.00  1066
## beta_edu[3] -0.9997 0.0718 -1.1435 -0.9978 -0.8627 1.00  1000
## rho          0.5259 0.1827  0.1648  0.5275  0.8518 1.00   743
## sd.theta     0.5808 0.0793  0.4394  0.5811  0.7466 1.00   871
## theta[1]     0.2257 0.4662 -0.6940  0.2215  1.1839 1.00  1015
## theta[2]    -1.0155 0.3190 -1.6820 -0.9943 -0.4337 1.00   874
## theta[3]    -0.9120 0.3447 -1.6465 -0.8976 -0.2913 1.01  1000
## theta[4]    -0.2419 0.4480 -1.1941 -0.2340  0.5680 1.01   982
## theta[5]    -0.2737 0.2797 -0.8320 -0.2766  0.2475 1.01  1074
## theta[6]    -0.2555 0.2700 -0.8135 -0.2487  0.2550 1.01   974
```

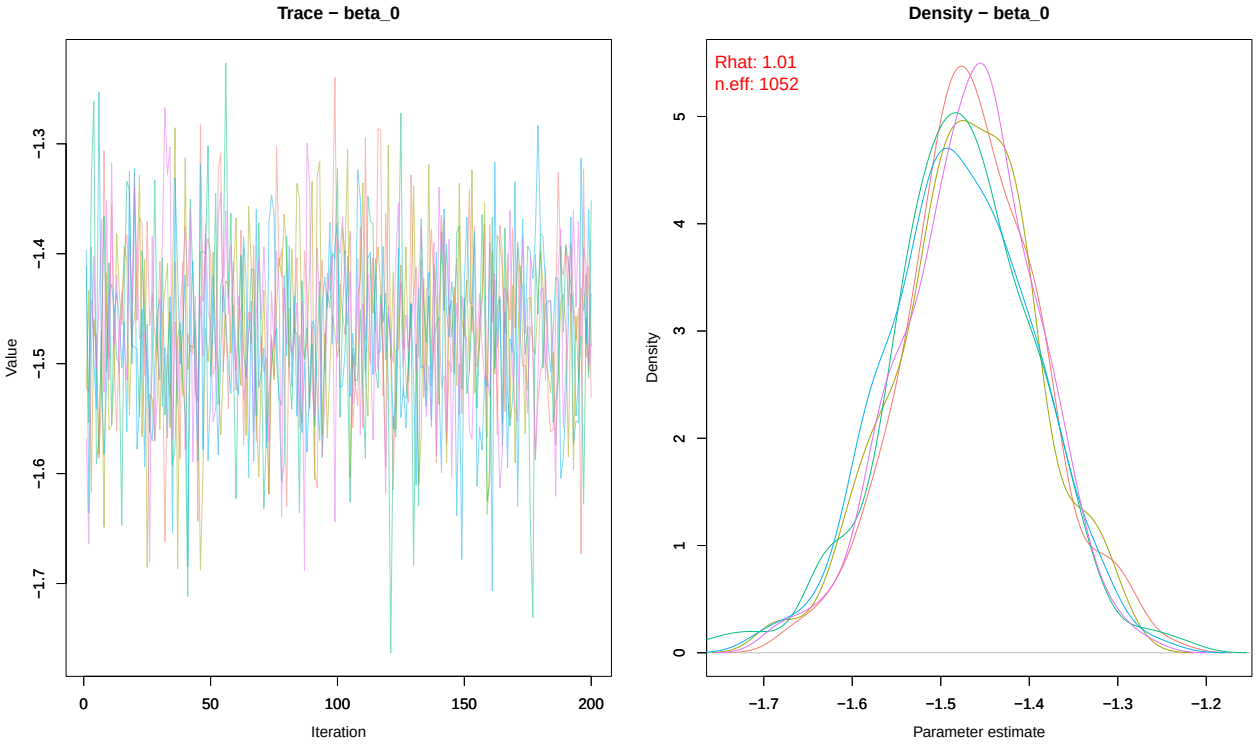
```
## theta[7]    -0.2560 0.3717 -0.9941 -0.2652  0.5050 1.01  2560
## theta[8]    -0.3068 0.3529 -0.9969 -0.3035  0.3778 1.00  1000
## theta[9]    -0.3275 0.4324 -1.2606 -0.3222  0.5251 1.00   957
## theta[10]   -0.1693 0.5154 -1.2063 -0.1646  0.8120 1.00  1086
## theta[11]   -0.7929 0.3097 -1.3880 -0.7781 -0.1850 1.00  1000
## theta[12]   -0.1473 0.3535 -0.8562 -0.1529  0.5692 1.00   941
## theta[13]   -0.2412 0.3309 -0.8529 -0.2291  0.3808 1.02  1257
## theta[14]    0.1602 0.3329 -0.4805  0.1645  0.8121 1.02   996
## theta[15]    0.2297 0.3540 -0.4425  0.2193  0.9387 1.00   855
## theta[16]   -0.0792 0.5656 -1.1448 -0.0817  1.0656 1.00  1003
## theta[17]   -0.3024 0.3688 -1.0388 -0.2884  0.3700 1.01   981
## theta[18]    0.2855 0.3800 -0.4396  0.2890  1.0721 1.00  1097
## theta[19]    0.0068 0.5371 -1.0292  0.0195  1.0342 1.00  1208
## theta[20]   -0.1700 0.4970 -1.1565 -0.1679  0.7960 1.02  1344
## theta[21]   -0.8612 0.3405 -1.5746 -0.8541 -0.2499 1.00   962
## theta[22]   -0.4802 0.2850 -1.0871 -0.4785  0.0393 1.00  1063
## theta[23]   -0.5720 0.3770 -1.3538 -0.5563  0.1340 1.01   884
## theta[24]   -0.4028 0.3811 -1.1022 -0.3986  0.3027 1.00  1000
## theta[25]   -0.8577 0.3467 -1.5467 -0.8447 -0.2068 1.00  1000
## theta[26]   -0.7418 0.4140 -1.6140 -0.7427  0.0480 1.00  1199
## theta[27]   -0.6845 0.5441 -1.7709 -0.6810  0.3750 1.01  1067
## theta[28]    1.0252 0.2793  0.5035  1.0059  1.6070 1.00   957
## theta[29]    0.7892 0.2654  0.3345  0.7722  1.3794 1.00  1036
## theta[30]    1.0309 0.5621 -0.0638  1.0287  2.1414 1.00  1000
## theta[31]    1.6047 0.4393  0.8028  1.5808  2.4716 1.00  1079
## theta[32]    0.4252 0.9863 -1.6240  0.4175  2.3601 1.00  1000
## theta[33]   -0.0470 0.6869 -1.4376 -0.0274  1.2387 1.01  1056
## theta[34]    0.8945 0.3298  0.2716  0.8993  1.5292 1.00   979
## theta[35]    0.8358 0.6290 -0.4006  0.8462  1.9801 1.00  1000
## theta[36]    0.8368 0.2732  0.3272  0.8219  1.4138 1.00   945
## theta[37]    0.9113 0.2201  0.5117  0.8944  1.3563 1.01  1000
## theta[38]    0.5875 0.4014 -0.2240  0.5826  1.3604 1.00  1170
## theta[39]    0.4860 0.7010 -0.8967  0.5004  1.7907 1.00   902
## theta[40]    0.7338 0.3831 -0.0058  0.7291  1.4718 1.00  1011
## theta[41]    1.5903 0.4699  0.6972  1.5780  2.5045 1.00   959
## theta[42]    0.4756 0.4500 -0.4063  0.4593  1.3805 1.00  1737
## theta[43]    1.1371 0.4639  0.2559  1.1248  2.1103 1.01  1186
## theta[44]    0.3340 0.3914 -0.4561  0.3404  1.1201 1.00  1175
## theta[45]    0.3050 0.6466 -1.0201  0.3121  1.5859 1.01  1197
## theta[46]    0.0453 0.6251 -1.2269  0.0539  1.2852 1.00  1000
## theta[47]   -0.1702 0.4582 -1.0492 -0.1610  0.7538 1.00  1064
## theta[48]    0.3970 0.5155 -0.6840  0.4021  1.4073 1.01  1047
## theta[49]    0.4364 0.6665 -0.8917  0.4514  1.6916 1.02  1272
## theta[50]    0.6948 0.4282 -0.1550  0.6888  1.5194 1.01  1011
## theta[51]    0.2993 0.3341 -0.3744  0.3053  0.9478 1.00   959
## theta[52]    0.4405 0.3118 -0.2124  0.4375  1.0620 1.00  1000
## theta[53]    0.0579 0.4275 -0.8011  0.0733  0.8795 1.00  1119
## theta[54]    0.0497 0.4729 -0.8707  0.0417  1.0103 1.00  1146
## theta[55]    0.1602 0.3189 -0.4605  0.1591  0.7749 1.01  1120
## theta[56]    0.6244 0.3433 -0.0022  0.6177  1.3044 1.00  1000
## theta[57]    0.6068 0.6173 -0.6430  0.5933  1.8496 1.01  1000
## theta[58]    0.1588 0.2898 -0.4096  0.1667  0.6992 1.00  1013
## theta[59]    0.4918 0.5265 -0.6005  0.4815  1.5358 1.00  1000
## theta[60]   -0.0557 0.3495 -0.7714 -0.0514  0.6393 1.01  1135
```

```
## theta[61]    0.8973 0.3806  0.1574  0.9027  1.6865 1.00   911
## theta[62]    0.7338 0.4395 -0.0846  0.7206  1.6334 1.01   922
## theta[63]    0.4667 0.4457 -0.3616  0.4623  1.2839 1.00  1053
## theta[64]    0.0339 0.5980 -1.1926  0.0296  1.1774 1.01  1174
## theta[65]    0.2447 0.5071 -0.7611  0.2406  1.2592 1.01  1086
## theta[66]    0.3443 0.4337 -0.5122  0.3437  1.1883 1.01   929
## theta[67]    0.2089 0.4795 -0.7581  0.2120  1.1502 1.00  1140
## theta[68]    0.8466 0.4668 -0.0510  0.8374  1.8518 1.01  1000
## theta[69]    0.7102 0.4635 -0.1617  0.7037  1.6136 1.01  1109
## theta[70]    0.9102 0.4206  0.0904  0.9069  1.7529 1.00  1052
## theta[71]    0.6942 0.4159 -0.1322  0.6983  1.5186 1.00  1000
## theta[72]   -0.1696 0.4582 -1.0889 -0.1479  0.7049 1.01  1060
## theta[73]    0.5875 0.4275 -0.2663  0.5845  1.3865 1.01  1000
## theta[74]    0.7838 0.5410 -0.2550  0.7638  1.8295 1.00  1121
## theta[75]    0.2930 0.5272 -0.8246  0.3054  1.2995 1.00  1298
## theta[76]    0.2074 0.4806 -0.7550  0.2028  1.1171 1.00  1384
## theta[77]    0.0285 0.4162 -0.7771  0.0205  0.8397 1.00  1261
## theta[78]    0.3863 0.4610 -0.5161  0.3869  1.3083 1.00  1352
## theta[79]    0.1500 0.3230 -0.4761  0.1459  0.7979 1.00  1040
## theta[80]   -0.0023 0.4599 -0.9283 -0.0147  0.8984 1.01   959
## theta[81]   -0.6024 0.1955 -1.0060 -0.5903 -0.2429 1.01   979
## theta[82]   -0.2704 0.2402 -0.7565 -0.2692  0.2195 1.01  1000
## theta[83]   -0.4514 0.3428 -1.1357 -0.4524  0.2166 1.00  1102
## theta[84]   -0.8876 0.2686 -1.4671 -0.8733 -0.3945 1.00  1000
## theta[85]    0.0233 0.2340 -0.4618  0.0268  0.4464 1.00   907
## theta[86]   -0.1633 0.5591 -1.2783 -0.1572  0.8729 1.00  1078
## theta[87]   -0.5611 0.4687 -1.4817 -0.5659  0.3517 1.00  1078
## theta[88]   -0.3763 0.4779 -1.3665 -0.3853  0.5478 1.00  1003
## theta[89]   -0.2631 0.3937 -1.0398 -0.2516  0.4921 1.00  1100
## theta[90]   -0.4023 0.3140 -1.0885 -0.3955  0.1994 1.00   905
## theta[91]   -0.3200 0.5214 -1.3391 -0.3334  0.7021 1.01  1052
## theta[92]   -0.6296 0.4317 -1.4729 -0.6141  0.2256 1.00  1017
## theta[93]   -0.1553 0.3054 -0.7906 -0.1417  0.4229 1.00  1647
## theta[94]   -0.0561 0.2952 -0.6213 -0.0580  0.5355 1.01  1000
## theta[95]   -0.4786 0.4729 -1.3700 -0.4705  0.4117 1.00  1036
## theta[96]   -0.3846 0.3398 -1.0892 -0.3951  0.2825 1.01  1202
## theta[97]   -0.0600 0.4131 -0.8780 -0.0682  0.7186 1.00  1198
## theta[98]    0.5910 0.2553  0.1045  0.5786  1.1012 1.00  1093
## theta[99]   -0.0571 0.6319 -1.3047 -0.0695  1.1954 1.00  1000
## theta[100]   1.0404 0.2970  0.5014  1.0253  1.6830 1.00   966
## theta[101]   0.2014 0.3058 -0.4070  0.1947  0.7932 1.01   962
## theta[102]   0.6828 0.4860 -0.2110  0.6946  1.6646 1.00  1023
```

```
MCMCsummary(object = bernoulli_results1$samples, params = "rho",
            # exact = TRUE,
            # ISB = FALSE,
            round = 4)
```

```
##       mean     sd   2.5%    50%  97.5% Rhat n.eff
## rho 0.5259 0.1827 0.1648 0.5275 0.8518    1   743
```

```
MCMCtrace(object = bernoulli_results1$samples,
          pdf = FALSE, # no export to PDF
          ind = TRUE, # separate density lines per chain
          Rhat = TRUE,
          n.eff = TRUE,
          params = "beta_0")
```




```
test <- bernoulli_results1$samples

which((MCMCsummary(object = test, params = "beta_0", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "beta_0", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "beta_age", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "beta_age", round = 4)[, 7] < 400))
```

```
## [1] 1
```

```
which((MCMCsummary(object = test, params = "beta_edu", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "beta_edu", round = 4)[, 7] < 400))
```

```
## [1] 1
```

```
which((MCMCsummary(object = test, params = "theta", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪  = test, params = "theta", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "sd.theta", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "sd.theta", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "rho", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪  = test, params = "rho", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
### Bernoulli (D-) model ###

round(bernoulli_results2$summary, 4)
```

```
##                mean     sd    2.5%     50%   97.5% Rhat n.eff
## beta_0      -3.4481 0.1811 -3.7970 -3.4525 -3.1059 1.00   843
## beta_age[1]  0.0000 0.0000  0.0000  0.0000  0.0000  NaN     0
## beta_age[2]  0.7014 0.1985  0.2953  0.6935  1.0706 1.01   732
## beta_age[3]  1.7697 0.1754  1.4279  1.7602  2.1191 1.00   948
## beta_edu[1]  0.0000 0.0000  0.0000  0.0000  0.0000  NaN     0
## beta_edu[2] -0.6625 0.1211 -0.8826 -0.6626 -0.4334 1.00  1351
## beta_edu[3] -1.3223 0.1454 -1.6033 -1.3241 -1.0282 1.00  1015
## rho          0.5285 0.1865  0.1752  0.5232  0.8698 1.00   840
## sd.theta     0.7771 0.1179  0.5648  0.7724  1.0342 1.00   879
## theta[1]    -0.8293 0.6407 -2.1496 -0.8290  0.3845 1.00   942
## theta[2]    -1.0446 0.4408 -1.9144 -1.0238 -0.2182 1.00  1000
## theta[3]     0.3286 0.3960 -0.4718  0.3354  1.0749 1.00  1040
## theta[4]    -0.0575 0.5315 -1.1610 -0.0405  1.0186 1.01   955
## theta[5]    -0.4865 0.4047 -1.3463 -0.4608  0.2731 1.00  1198
## theta[6]    -0.4813 0.3937 -1.2697 -0.4669  0.2549 1.00  1002
## theta[7]     0.3027 0.4184 -0.5401  0.3026  1.0738 1.01  1109
## theta[8]     0.1814 0.4004 -0.6316  0.1971  0.9164 1.01   870
## theta[9]     0.4192 0.4832 -0.5211  0.4405  1.4011 1.00   956
## theta[10]    0.1196 0.6604 -1.2227  0.1136  1.3261 1.00  1000
## theta[11]   -0.4269 0.3938 -1.2537 -0.4296  0.3535 1.00  1044
## theta[12]    0.0421 0.4270 -0.8215  0.0532  0.8754 1.00  1220
## theta[13]   -0.1365 0.4066 -0.9488 -0.1280  0.6314 1.00  1000
## theta[14]    0.2985 0.3822 -0.4574  0.3057  1.0313 1.00  1026
## theta[15]   -0.1870 0.4950 -1.2368 -0.1892  0.7897 1.00  1014
## theta[16]    0.3152 0.6186 -0.8775  0.3221  1.5108 1.00  1072
## theta[17]    0.3389 0.4099 -0.4512  0.3298  1.1325 1.00  1256
## theta[18]    0.4506 0.4072 -0.3808  0.4399  1.2222 1.00  1195
## theta[19]    0.1798 0.5759 -0.9665  0.1970  1.2962 1.00  1000
```

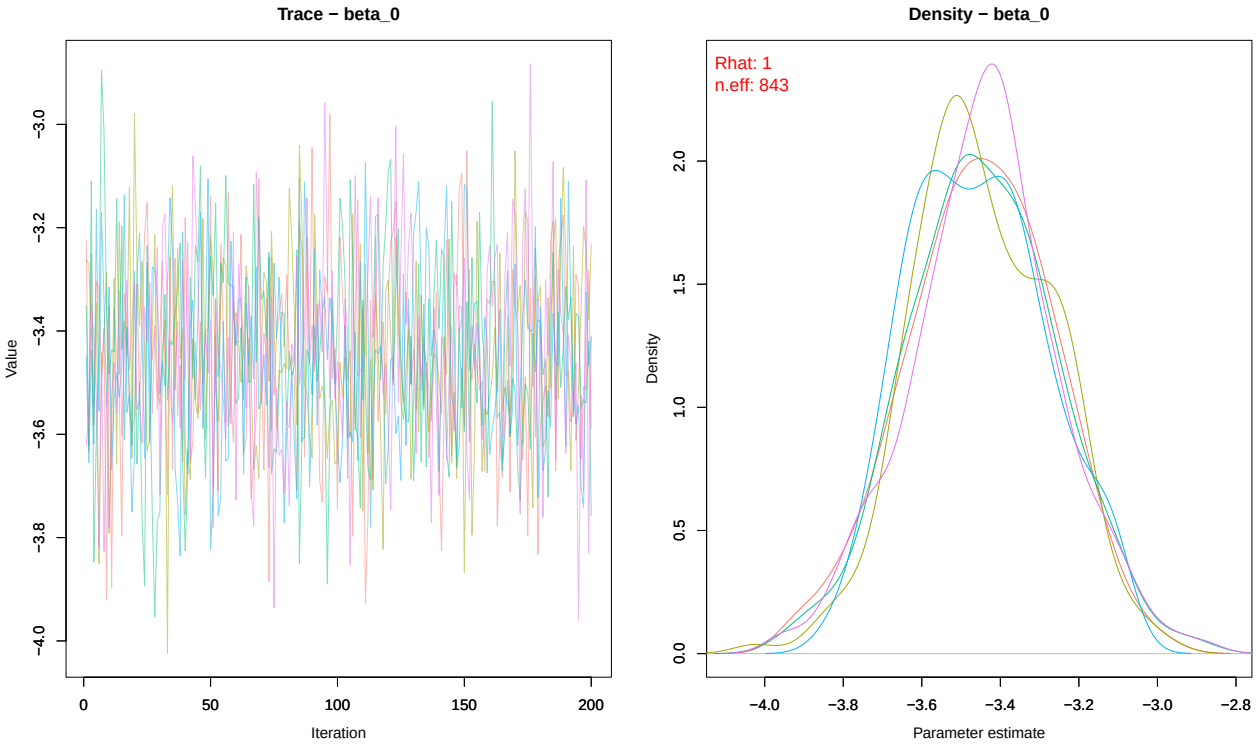
```
## theta[20]    0.3498 0.5313 -0.7664  0.3655  1.3665 1.00  1016
## theta[21]   -0.3744 0.4197 -1.2201 -0.3636  0.4247 1.00  1218
## theta[22]   -0.5459 0.4138 -1.3758 -0.5467  0.2240 1.01  1125
## theta[23]   -0.2586 0.4946 -1.2640 -0.2540  0.7120 1.00  1116
## theta[24]   -0.0960 0.4885 -1.0860 -0.0842  0.7879 1.01  1000
## theta[25]   -0.2866 0.3910 -1.0912 -0.2697  0.4537 1.00   954
## theta[26]   -0.5420 0.4886 -1.4713 -0.5398  0.4043 1.00   965
## theta[27]   -0.5467 0.6370 -1.8124 -0.5295  0.7013 1.00  1129
## theta[28]    1.1509 0.3195  0.5731  1.1388  1.8269 1.01  1000
## theta[29]    1.1178 0.2922  0.5429  1.1076  1.7351 1.00   938
## theta[30]    0.8922 0.6514 -0.4329  0.9283  2.1361 1.01   923
## theta[31]    1.1518 0.4757  0.2717  1.1452  2.0906 1.00  1013
## theta[32]    0.3216 1.0476 -1.6732  0.3314  2.3534 1.01   928
## theta[33]    0.3856 0.7294 -1.0998  0.4000  1.7646 1.00  1158
## theta[34]    1.2736 0.3618  0.5843  1.2808  2.0148 1.00  1046
## theta[35]    0.8200 0.6634 -0.5452  0.8133  2.0504 1.00  1000
## theta[36]    0.6275 0.3184 -0.0544  0.6134  1.2651 1.00  1140
## theta[37]    0.8735 0.2614  0.3888  0.8659  1.4289 1.00  1000
## theta[38]    0.8164 0.4382 -0.0090  0.7872  1.7282 1.00   993
## theta[39]   -0.0204 0.7473 -1.4362 -0.0413  1.4560 1.00   830
## theta[40]    1.0496 0.4415  0.1742  1.0624  1.8919 1.00  1216
## theta[41]    1.4862 0.4836  0.5817  1.4925  2.4380 1.00  1110
## theta[42]    0.1399 0.5769 -1.0070  0.1554  1.2394 1.00  1193
## theta[43]    1.6677 0.4768  0.7565  1.6773  2.6542 1.00  1072
## theta[44]    0.3997 0.4441 -0.4773  0.3978  1.2891 1.01  1000
## theta[45]    0.0274 0.7216 -1.4084  0.0436  1.4042 1.00  1000
## theta[46]    0.3573 0.6541 -0.9142  0.3588  1.6466 1.00  1561
## theta[47]   -0.0334 0.5245 -1.0408 -0.0254  0.9597 1.01   960
## theta[48]    0.3036 0.5658 -0.8422  0.3050  1.3770 1.01   956
## theta[49]    0.1296 0.6832 -1.2389  0.1311  1.4655 1.02  1030
## theta[50]    0.4329 0.4628 -0.4970  0.4228  1.3776 1.02   954
## theta[51]   -0.1711 0.4883 -1.2900 -0.1418  0.7391 1.01  1099
## theta[52]    0.3012 0.3558 -0.4060  0.3003  1.0031 1.00  1000
## theta[53]   -0.0141 0.4550 -0.9375  0.0209  0.8453 1.01  1074
## theta[54]    0.6699 0.4842 -0.2534  0.6750  1.6652 1.00  1131
## theta[55]   -0.2364 0.4011 -1.0398 -0.2221  0.5152 1.01   847
## theta[56]    0.0595 0.4024 -0.7213  0.0703  0.7891 1.00  1000
## theta[57]   -0.4503 0.7259 -1.9334 -0.4247  0.8697 1.01   943
## theta[58]    0.4190 0.3397 -0.2579  0.4141  1.0664 1.00  1031
## theta[59]    0.4258 0.5662 -0.7042  0.4444  1.5206 1.00   946
## theta[60]    0.5285 0.3908 -0.2248  0.5352  1.2997 1.00  1000
## theta[61]    0.0362 0.4627 -0.8912  0.0524  0.9533 1.00  1047
## theta[62]    0.3547 0.4933 -0.5997  0.3643  1.3430 1.00  1778
## theta[63]    0.0899 0.4998 -0.8697  0.0958  1.0712 1.00  1132
## theta[64]   -0.1514 0.6398 -1.4139 -0.1408  1.1238 1.01  1056
## theta[65]    0.6703 0.5425 -0.3585  0.6602  1.7102 1.01   969
## theta[66]    0.6993 0.4843 -0.3325  0.7115  1.5991 1.00   957
## theta[67]    0.4697 0.5368 -0.6095  0.4744  1.4859 1.00  1058
## theta[68]    0.3598 0.5104 -0.6854  0.3686  1.2715 1.00  1166
## theta[69]    0.3725 0.4550 -0.5855  0.3787  1.2424 1.00  1056
## theta[70]    0.2549 0.4915 -0.7042  0.2589  1.2051 1.01  1000
## theta[71]    0.2755 0.4897 -0.7232  0.3048  1.1967 1.00  1017
## theta[72]   -0.5222 0.6185 -1.7841 -0.5148  0.6388 1.00  1043
## theta[73]   -0.1340 0.5026 -1.1985 -0.1428  0.8744 1.00  1102
```

```
## theta[74]    0.3238 0.5657 -0.7442  0.3027  1.4904 1.00  1000
## theta[75]   -0.0473 0.5957 -1.4318 -0.0318  1.0390 1.00  1000
## theta[76]    0.2083 0.5139 -0.7675  0.2283  1.1904 1.00   958
## theta[77]   -0.1059 0.4571 -0.9996 -0.1001  0.7254 1.01  1000
## theta[78]    0.0107 0.5125 -1.0021  0.0025  1.0305 1.00  1107
## theta[79]    0.0674 0.3865 -0.6862  0.0704  0.8493 1.00  1126
## theta[80]    0.3563 0.5195 -0.7224  0.3797  1.3042 1.00  1531
## theta[81]   -0.6913 0.2865 -1.2768 -0.6808 -0.1675 1.00  1006
## theta[82]   -0.3600 0.3228 -1.0260 -0.3335  0.2499 1.00  1000
## theta[83]   -0.8193 0.5242 -2.0328 -0.7649  0.0705 1.00  1010
## theta[84]   -0.9182 0.3622 -1.6906 -0.8993 -0.2813 1.00   955
## theta[85]   -0.3664 0.3360 -1.0850 -0.3576  0.2568 1.01  1000
## theta[86]   -0.3130 0.6531 -1.6242 -0.3352  0.9130 1.01  1050
## theta[87]   -0.0846 0.5153 -1.0589 -0.0812  0.9130 1.00   954
## theta[88]   -0.0079 0.5433 -1.1515  0.0093  1.0380 1.01  1066
## theta[89]   -0.2217 0.4743 -1.1635 -0.2108  0.6568 1.00  1207
## theta[90]    0.0916 0.3736 -0.6680  0.0955  0.8295 1.01  1000
## theta[91]   -0.2683 0.5553 -1.3812 -0.2753  0.8539 1.00  1065
## theta[92]   -0.3365 0.5262 -1.4463 -0.3292  0.6173 1.01  1046
## theta[93]   -0.0577 0.4103 -0.9029 -0.0311  0.6771 1.00  1000
## theta[94]   -0.0324 0.3633 -0.7649 -0.0189  0.6387 1.00  1067
## theta[95]   -0.0554 0.5408 -1.1108 -0.0579  1.0539 1.00  1161
## theta[96]   -0.1102 0.4039 -0.9549 -0.1084  0.6928 1.00   969
## theta[97]   -0.0741 0.4684 -1.0179 -0.0499  0.8348 1.01  1093
## theta[98]   -0.1566 0.3206 -0.8346 -0.1527  0.4272 1.01  1064
## theta[99]   -0.3195 0.6932 -1.6784 -0.3087  0.9829 1.00  1000
## theta[100]  -0.0722 0.3346 -0.7507 -0.0643  0.5685 1.00  1005
## theta[101]   0.1070 0.3707 -0.6485  0.1276  0.7876 1.01  1151
## theta[102]   0.5827 0.4665 -0.3532  0.5837  1.4723 1.00  1087
```

```
MCMCsummary(object = bernoulli_results2$samples, params = "rho",
            # exact = TRUE,
            # ISB = FALSE,
            round = 4)
```

```
##       mean     sd   2.5%    50%  97.5% Rhat n.eff
## rho 0.5285 0.1865 0.1752 0.5232 0.8698    1   840
```

```
MCMCtrace(object = bernoulli_results2$samples,
          pdf = FALSE, # no export to PDF
          ind = TRUE, # separate density lines per chain
          Rhat = TRUE,
          n.eff = TRUE,
          params = "beta_0")
```




```
test <- bernoulli_results2$samples

which((MCMCsummary(object = test, params = "beta_0", round = 4)[, 6]) > 1.02 |
↪   (MCMCsummary(object = test, params = "beta_0", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "beta_age", round = 4)[, 6]) > 1.02 |
↪   (MCMCsummary(object = test, params = "beta_age", round = 4)[, 7] < 400))
```

```
## [1] 1
```

```
which((MCMCsummary(object = test, params = "beta_edu", round = 4)[, 6]) > 1.02 |
↪   (MCMCsummary(object = test, params = "beta_edu", round = 4)[, 7] < 400))
```

```
## [1] 1
```

```
which((MCMCsummary(object = test, params = "theta", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪   = test, params = "theta", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "sd.theta", round = 4)[, 6]) > 1.02 |
↪  (MCMCsummary(object = test, params = "sd.theta", round = 4)[, 7] < 400))
```

```
## integer(0)
```

```
which((MCMCsummary(object = test, params = "rho", round = 4)[, 6]) > 1.02 | (MCMCsummary(object
↪  = test, params = "rho", round = 4)[, 7] < 400))
```

```
## integer(0)
```